

SCSR1033
Computer Organization and Architecture

Lecture slides prepared by “Computer Organization and Architecture”, 9/e, by William Stallings, 2013.

Welcome

SCSR1033 – Computer Organization & Architecture

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THE MODULES

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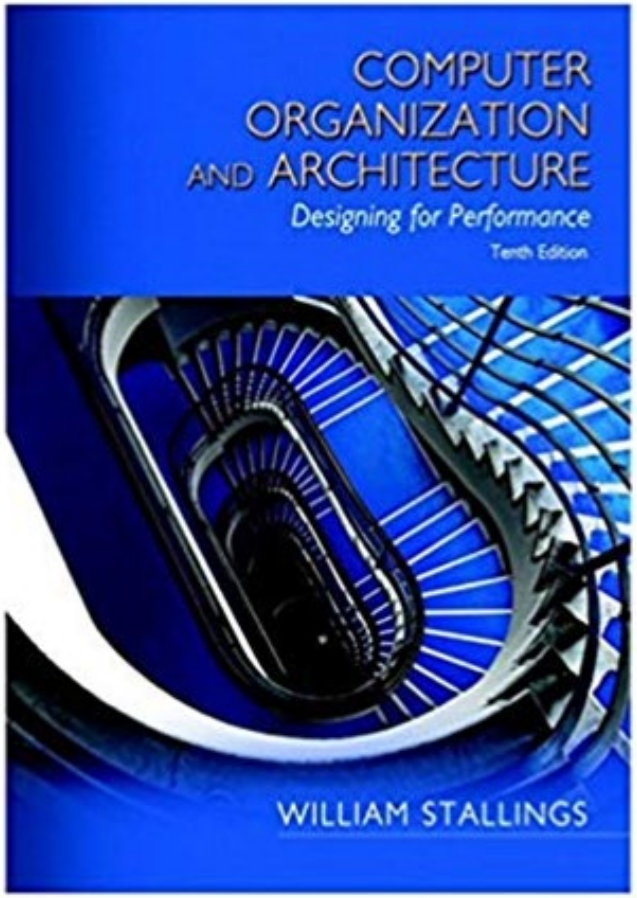
Computer Organization and Architecture (SCSR1033) will cover the following modules:

- Module 1: Introduction
- Module 2: Data Representations
- Module 3: Introduction to Assembly Language
- Module 4: Instruction Set Architecture (ISA)
- Module 5: CPU
- Module 6: Memory
- Module 7: Input/Output
- Module 8: Computer Performance

THE TEXTBOOK

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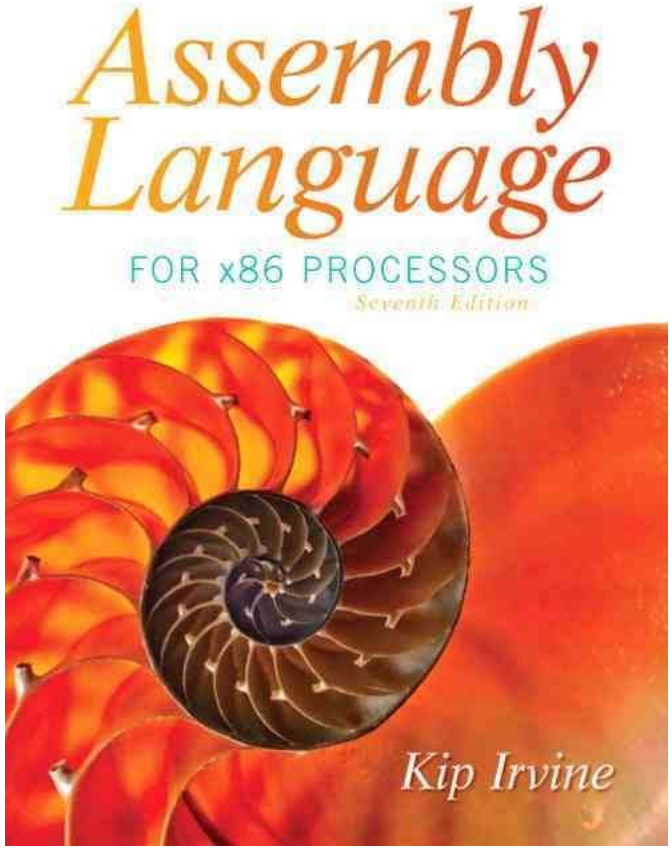
Computer Organization and Architecture, 10th Edition (2016)
William Stallings



THE TEXTBOOK

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Assembly Language for x86 Processors, 7th Edition (2015)
Kip R. Irvine, Florida International University



THE SOFTWARE

1

MS Visual Studio 2019

For lab exercises



Grading



Continuous Assessment		Percentage
1	Labs	15
2	Quiz (4)	20
3	Mid Term	20
4	Group Project	15
Final Assessment		Percentage
1	Final Examination	30

UTM
UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

SCSR1033 COA

Special requirement to deliver the course BSE Assessment starting Sem II 2024/25

PLEASE UNDERSTAND THE NEW REQUIREMENT FOR PASSING MARK (SCSR1033)

All students must adhere to the following to pass a computing course:

i. **Pass both** continuous assessment **AND** final assessment

AND

ii. Get at least **minimum pass marks** following UTM regulation (**40%**)

- The passing mark for Continuous Assessment is **30%** of coursework marks.
- The passing mark for the Final Assessment is **20%** of the final exam marks.

BSE Assessment starting Sem II 2024/25

Continuous Assessment (70%)		Percentage
1	Labs (1-3)	15
2	Quiz (1-4)	20
3	Mid Term	20
4	Group Project	15
Final Assessment (30%)		
1	Final Examination	30
	Total	100

Simulation

- The passing mark for Continuous Assessment is 30% of continuous marks.
- The passing mark for the Final Assessment is 20% of the final marks.

	CA (70%) Continuous Assesment (70%)	Achieved CA (%)	Meet Req (CA)	Final (100%)	Final (30%)	Achieved Final (%)	Meet Req (Final)	Total (100)	Req (pass UTM 40%)	Grade	Reason
Student 1	21	30	Y	20	6	20	Y	27	N	Fail	Not Meet Req UTM (at least 40% of Total)
Student 2	50	71	Y	10	3	10	N	53	Y	Fail	Not Meet Req Final (at least 20%)
Student 3	40	57	Y	30	9	30	Y	49	Y	Pass	Meet all Req
Student 4	15	21	N	90	27	90	Y	42	Y	Fail	Not Meet Req CA (at least 20%)
Student 4	60	86	Y	80	24	80	Y	84	Y	Pass	Meet all Req

Mark your calendar



- **Mid Term Test:**
  - **Date:**
    - 21 May 2025 (Wed)
    - ~ Week 10
  - **Time:**
    - 8 pm – 10 pm
  - **Coverage:**
    - Chap 1 - 4

**Please make sure you put these dates into your calendar, and be ready for the tests

**Please ensure there is no 'double bookings'

Module 1
Introduction

Objectives:

- To study of computers with a brief history that serves the purpose of providing an overview of computer structure and functions.
- To differentiate between computer architecture and computer organization.
- To understand what makes a computer system tick before attempt to optimize the programs that it runs.

Module 1
Introduction



1.1 An Overview
1.2 Components of Computer
1.3 Computer Structure and Functions
1.4 Computer Evolution
1.5 Computer Level Hierarchy
1.6 An Example System
1.7 Summary

Module 1
Introduction

1.1 An Overview

- The Computer Family
- Computer Types

1.2 Components of Computer
1.3 Computer Structure and Functions
1.4 Computer Evolution
1.5 Computer Level Hierarchy
1.6 An Example System
1.7 Summary

1.1 An Overview

1

- This subject is about the structure and function of computers in order to differentiate the nature and characteristics of modern-day computers clearly.



- This task is challenging because:

  1) Variety of computer products from a single-chip microcomputers to supercomputers.

     - Variety not only in cost, but also in size, performance, and application.

  2) Rapid changes in all aspect that characterized computer technology continuously.

- In describing computers, a distinction is often made between computer architecture and computer organization.

Computer architecture	Computer organization
Refer to those attributes of a system: • that visible to a programmer • Have a direct impact on the logical execution of a program.	Refer to the operational units and their interconnections that realize the architectural specification.
How do I design a computer?	*How does a computer work?*

William Stallings (2013). *Computer Organization and Architecture: Designing for Performance* (9th Edition). United States: Pearson Education Limited, p.29.
Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.2.

1

- Computer architecture and organization include many elements to be addressed:

Computer architecture	Computer organization
• *Instruction sets & formats.* • *Operation codes.* • *Data types* • *Number & types of registers* • *Addressing modes.* • *Memory access methods.* • *I/O mechanisms.*	• *Control signal* • *Signaling method* • *Memory types.*

1 The Computer Family

- Many computer manufacturers offer a family of computer models, all with the same architecture but with differences in organization.
- All Intel x86 family share the same basic architecture.
- The IBM System/370 architecture first introduced in 1970 included a number of models that share the same basic architecture and has survived to this day as the architecture of IBM's mainframe product line.
- The newer models retained the same architecture so that the customer's software investment was protected (code compatibility).

The computer family

1

S/370 replacement [edit]

The System/370 line was replaced by the IBM System/390 in the 1990s, and the architecture was similarly renamed from ESA/370 to ESA/390. This was essentially just a rename for marketing reasons, rather than major architectural change.[*citation needed*]

In 2000, the System/390 was replaced by the zSeries (now called IBM System z). The zSeries mainframes introduced the 64-bit z/Architecture, the most significant design improvement since the 31-bit transition.[*citation needed*] All have retained essential backward compatibility with the original S/360 architecture and instruction set.

IBM S/370 (mainframe, year 1970)

1



IBM zSeries (mainframe, year 2017)

1



Same architecture BUT differences in organization !





1933 vs. 1948 Chevrolet

From whittling a 2x4 to a smooth clay styling model





1948 vs. 1963 Ford

From fat fendered to smooth, integrated design





1963 vs. 1978 Volkswagen

From 1930s Hitler to 1970s sharp-edged Giugiaro





1978 vs. 1993 Lincoln

From brick to jelly bean





1993 vs. 2008 Toyota

Miniscule style evolution

www.joesherlock.com

Proton Saga (generations)

1



- The generations of the Proton Saga
- https://en.wikipedia.org/wiki/Proton_Saga

Same architecture
BUT
differences in
organization !

Honda Civic (generations)

1

5th 1991 - 1995
CIVIC
"Sport Civic" จากแนวคิดพื้นฐาน ของการออกแบบ
รถยนต์ สู่ยนตรกรรมระดับโลก



6th 1995 - 2000
CIVIC
"Miracle Civic" เส้นสายแข็งแกร่งสร้างงาม เปิดพลัง
ออกแบบตามหลักอากาศพลศาสตร์



7th 2000 - 2005
CIVIC
จากการพัฒนาด้านความปลอดภัยอย่างจริงจัง โดย
ฮอนด้าได้สร้างศูนย์ทดสอบการชนในร่ม
(Real World Crash Test Facilities)



8th 2005 ~
CIVIC
ความล้ำหน้าทางดีไซน์และเทคโนโลยี คือหัวใจของ
ซีวิค จากรุ่นสู่รุ่น



1st 1972 - 1979
CIVIC
ฮอนด้า ซีวิค รุ่นแรก เปิดตัวเมื่อปี ค.ศ.1972
(พ.ศ.2515) ด้วยเครื่องยนต์ CVCC



2nd 1979 - 1983
CIVIC
"SuperCivic" ได้รับความนิยมอย่างกว้างขวาง
ด้วยรูปทรง 2 มิติ (two-box styling)แบบมี
เอกลักษณ์



3rd 1983 - 1987
CIVIC
"Wonder Civic" รับสนุก เร้าใจ พร้อมห้องเก็บ
สัมภาระท้ายกว้างขวางภายใต้แนวคิดM/M เพื่อเพิ่ม
พื้นที่สำหรับผู้ใช้รถยนต์



4th 1987 - 1991
CIVIC
"Grand Civic" ใช้ระบบกันสะเทือนแบบอิสระ 4 ล้อ



Same architecture
BUT
differences in
organization !

Honda Civic (generations)

1



Same architecture
BUT
differences in
organization !



One advantage having the SAME architecture is that the same software can be used in the newer computer models with DIFFERENT computer organization.

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Why study computer architecture and computer organization?



- To design better programs, including system software such as compilers, operating systems, and device drivers.
- To optimize program behavior.
- To evaluate (benchmark) computer system performance.
- To understand time, space, and price tradeoffs.
  - *e.g.* Solving calculation in less time but more memory.

Computer Science Curricula (ACM/IEEE)

1

Architecture and Organization (AR)

Computing professionals should not regard the computer as just a black box that executes programs by magic. The knowledge area Architecture and Organization builds on Systems Fundamentals (SF) to develop a deeper understanding of the hardware environment upon which all computing is based, and the interface it provides to higher software layers. Students should acquire an understanding and appreciation of a computer system's functional components, their characteristics, performance, and interactions, and, in particular, the challenge of harnessing parallelism to sustain performance improvements now and into the future. Students need to understand computer architecture to develop programs that can achieve high performance through a programmer's awareness of parallelism and latency. In selecting a system to use, students should be able to understand the tradeoff among various components, such as CPU clock speed, cycles per instruction, memory size, and average memory access time.

1 Computer Types

- Computers can be generally classified by size and power as follows, though there is considerable overlap:



- **Personal Computer (PC)** : A small, single-user computer based on a microprocessor.



- **Workstation** : A powerful, single-user computer. A workstation is like a PC, but it has a more powerful microprocessor and, in general, a higher-quality monitor.

https://www.syniaditsolutions.co.uk/wp-content/uploads/2017/12/computer-HAM.jpg | https://www.dpreview.com/files/p/articles/4907218890/HPZ8.jpeg

- **Minicomputer** : A multi-user computer capable of supporting up to hundreds of users simultaneously.



- **Mainframe** : A powerful multi-user computer capable of supporting many hundreds or thousands of users simultaneously.



http://3.bp.blogspot.com/-9L_4vswIwVM/Tc5XGFZ23rI/AAAAAAAADJI/4n9pU3lxF64/s1600/ibm+as+400.jpg
https://images.idgesg.net/images/article/2017/07/ibm_z_1-100729283-large.jpg

- **Supercomputer** :
  An extremely fast computer that can perform hundreds of millions of instructions per second.



THE WORLD'S FASTEST SUPERCOMPUTER IS BACK IN AMERICA

Meet Summit

By Micah Singleton | @MicahSingleton | Jun 12, 2018, 4:35pm EDT



- **Mobile Devices** :

  A computing device that small enough to hold and operate in the hand (like smartphones, tablets, smart watches etc).

https://cdn.vox-cdn.com/thumbor/58Z616icwM8mzzDZo76p3iHSZNo=/0x0:2040x1360/2320x1305/filters:focal(661x498:987x824)/cdn.vox-cdn.com/uploads/chorus_image/image/60039409/msingleton_180612_2663_0006.0.jpg
http://www.northstarbusinessessentials.com/wp-content/uploads/2015/10/Mobile-devices.jpg

Module 1
Introduction

1.1 An Overview
1.2 Components of Computer
1.3 Computer Structure and Function
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1.5 Computer Level Hierarchy
1.6 An Example System
1.7 Summary

1.2 Components of Computer

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- **Computers : Definition**

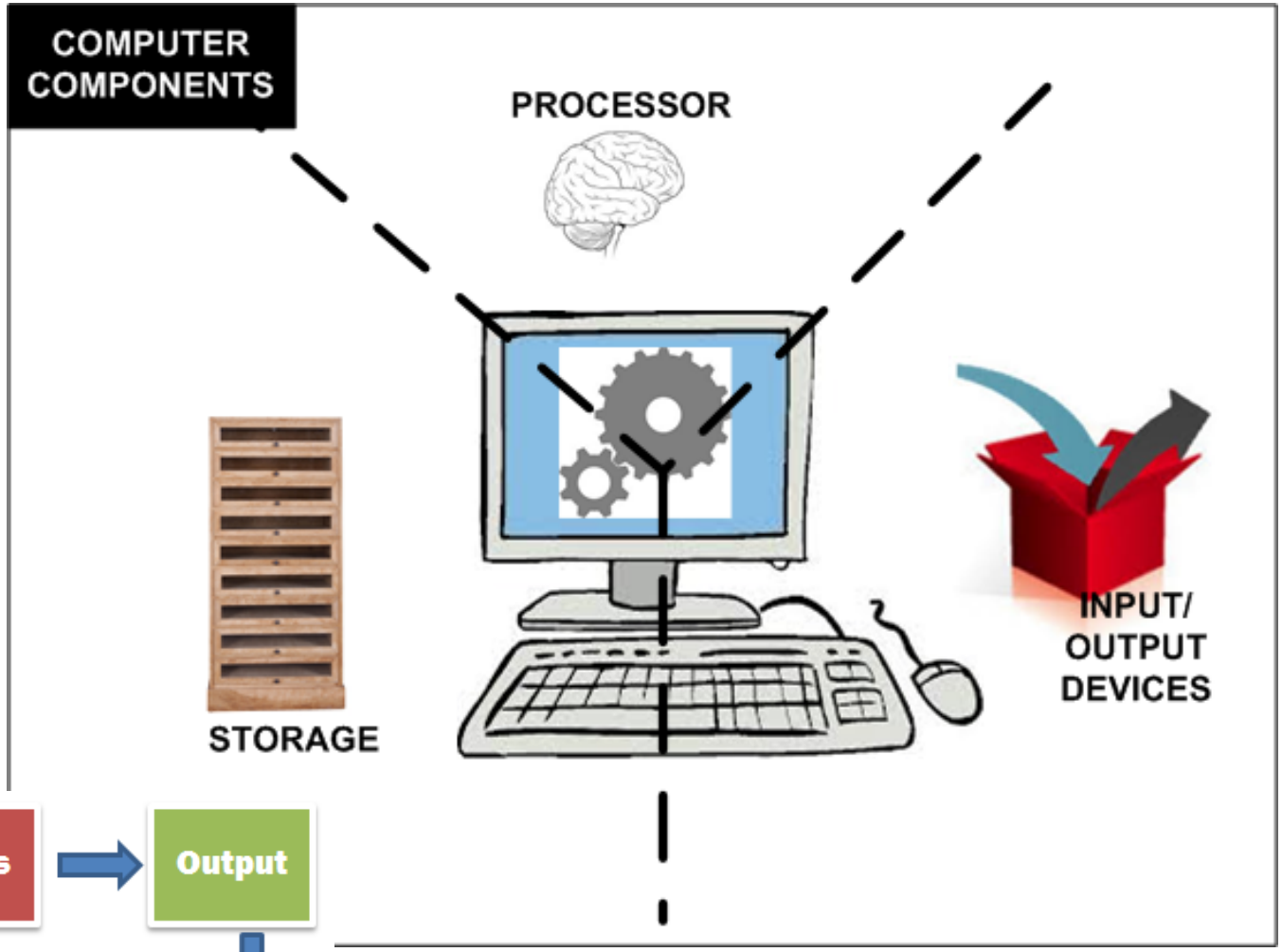


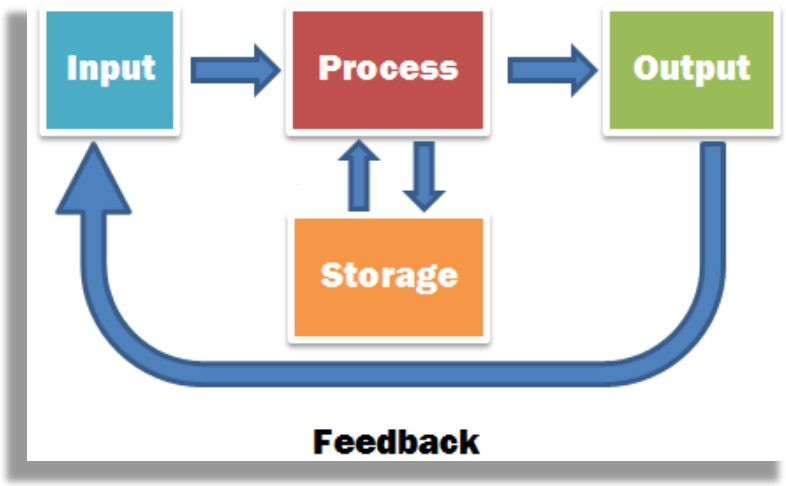
- An electronic device for storing and processing data, typically in binary form, according to instructions given to it in a variable program.

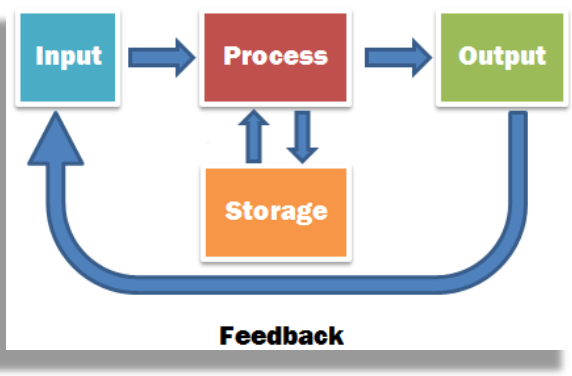
- An electronic device designed to accept data, perform prescribed mathematical and logical operations at high speed, and display the results of these operations.

Keywords: *store*, *process*, *input/output (I/O)*

https://media.proprofs.com/images/QM/user_images/1752848/qm6146054691.png


COMPUTER COMPONENTS
PROCESSOR
STORAGE
INPUT/ OUTPUT DEVICES


Input
Process
Output
Storage
Feedback



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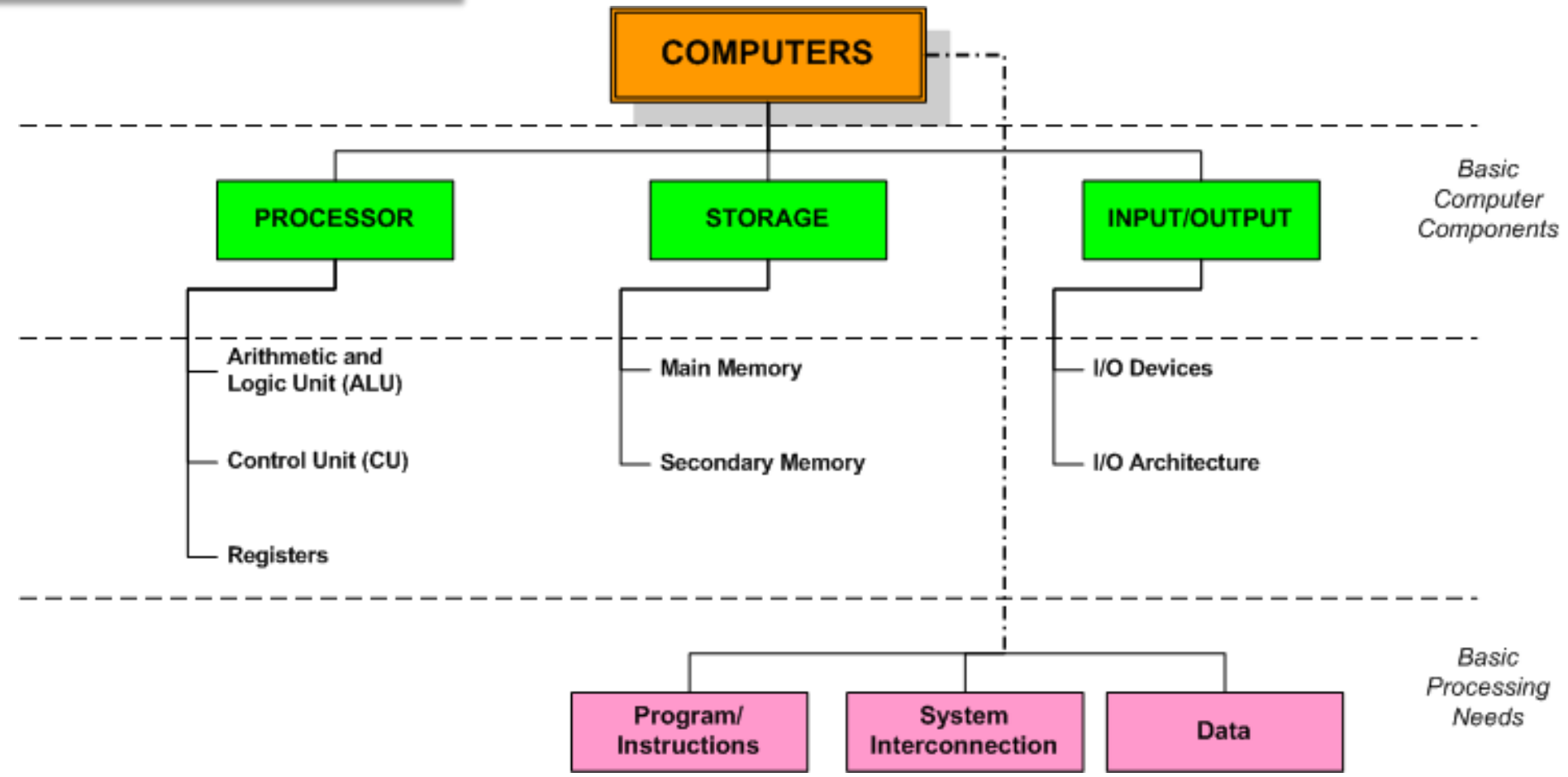


Figure: Computer components in detail

Module 1
Introduction

1.1 An Overview

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1.6 An Example System

1.7 Summary

- Overview
- Structure
- Functions
- Program

1.3 Computer Structure & Functions

1

- A computer is a complex system with a hierarchical system of interrelated subsystems with different levels.
- At each level, the designer is concerned with structure and function:

Structure	Functions
The way in which the components are interrelated.	The operation of each individual component as part of the structure.

CPU (Control Processing Unit)
I/O (Input/Output)
ALU (Arithmetic Logic Unit)

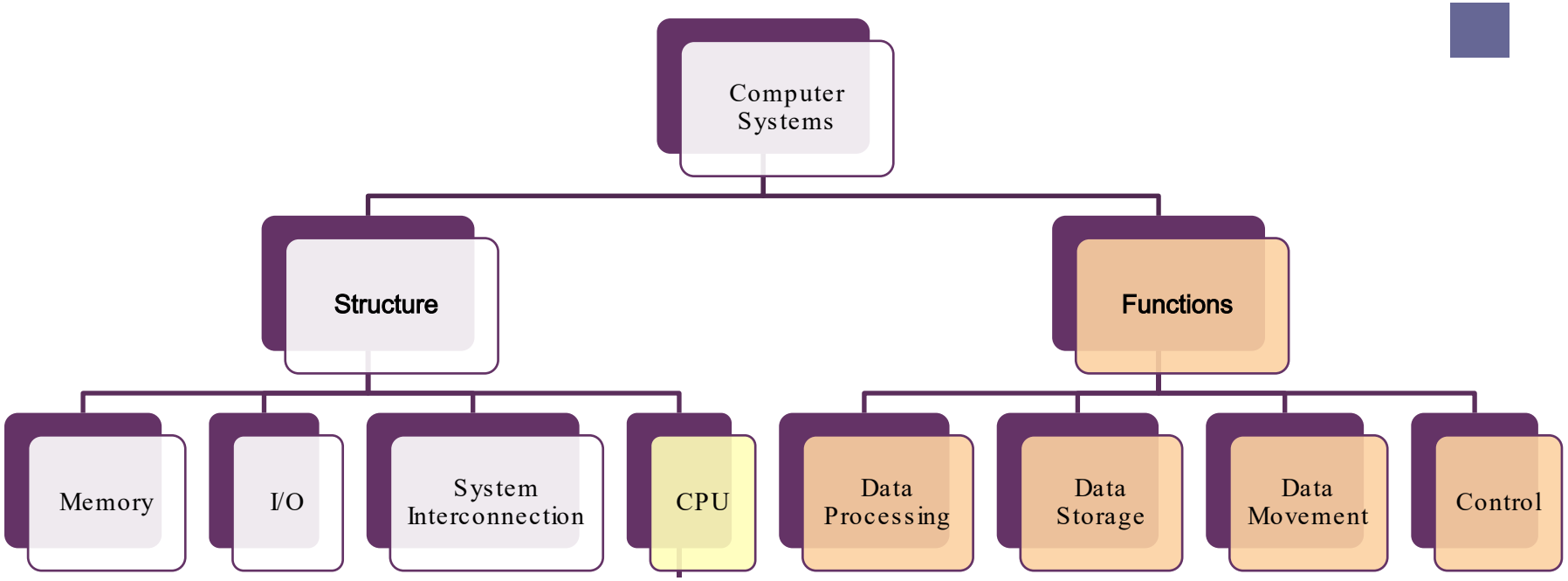
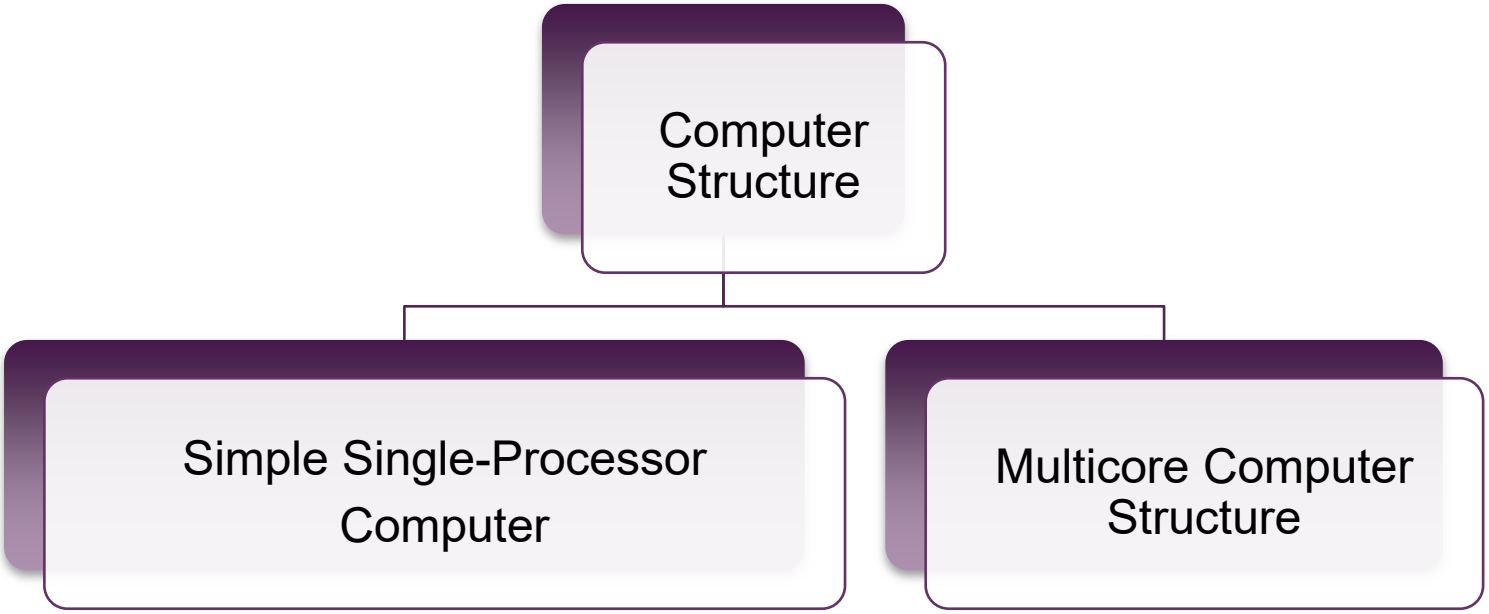


Figure : Computer components in each level

1 Structure

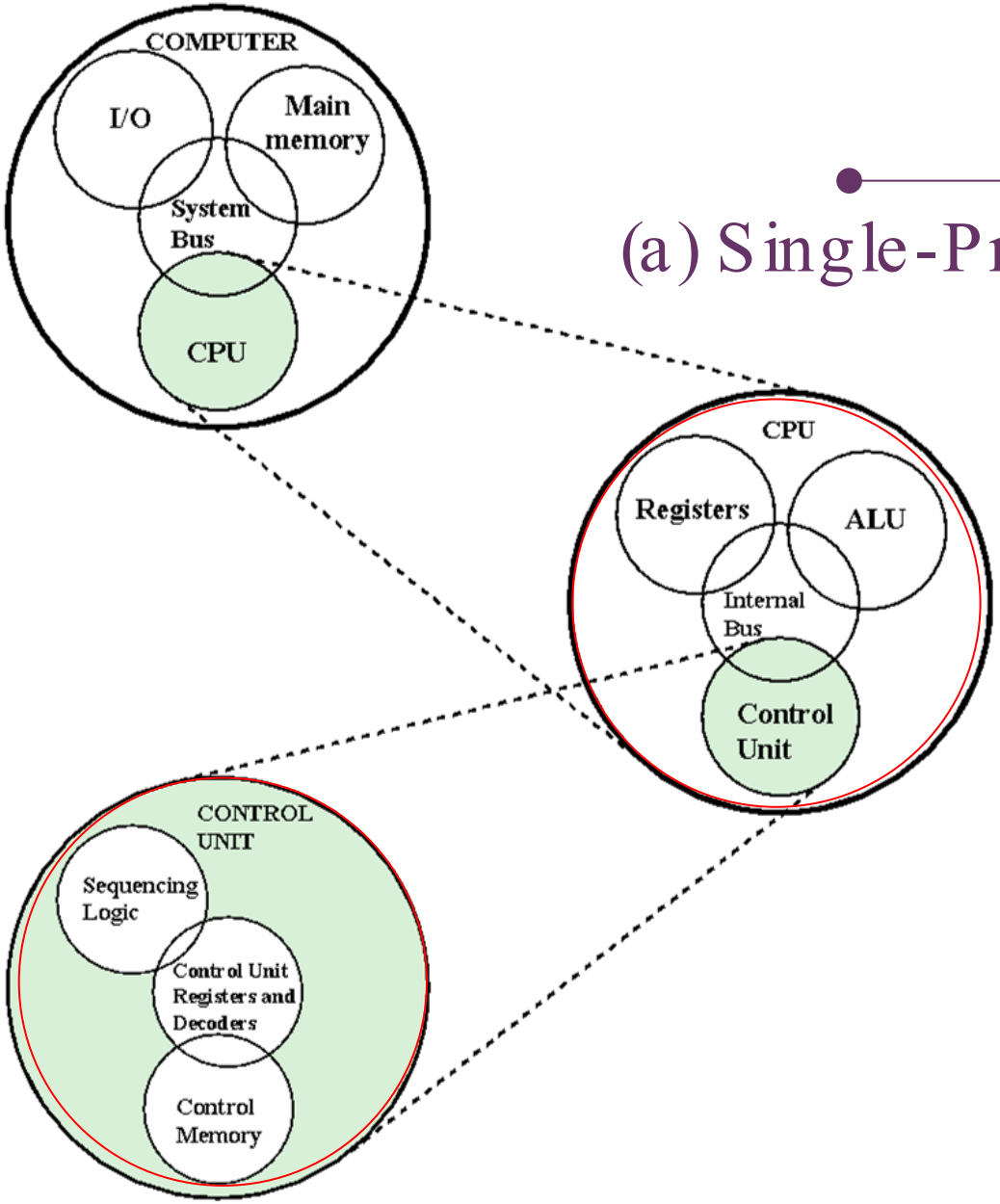
- General category of the internal structure of computer:



(Traditional Computer)

1 Structure

(a) Single-Processor Computer



- **Figure :** Simplified view of major elements for single core computer

Table: The task for each structural component.

Components	Task
CPU *(Processor)*	Controls the operation of the computer and performs its data processing functions.
Memory	Stores data.
I/O	Moves data between the computer and its external environment.
System Interconnection *(System bus)*	Mechanism for communication among CPU, main memory, and I/O.

Table: The major structural components of CPU.

Components	Task
ALU	Performs the computer's data processing functions.
Control Unit (CU)	Controls CPU operations.
Registers	Provides internal storage to the CPU.
CPU Interconnection	Mechanism for communication among the ALU, Control Unit, and Registers.

1 Structure

(b) Multicore Computer Structure

- Generally, the contemporary computers have multiple processors.
- All processors reside on s single chip.
- Each processing units (CU, ALU, registers) is called a core.
- The use of multiple layers of memory is called cache memory.

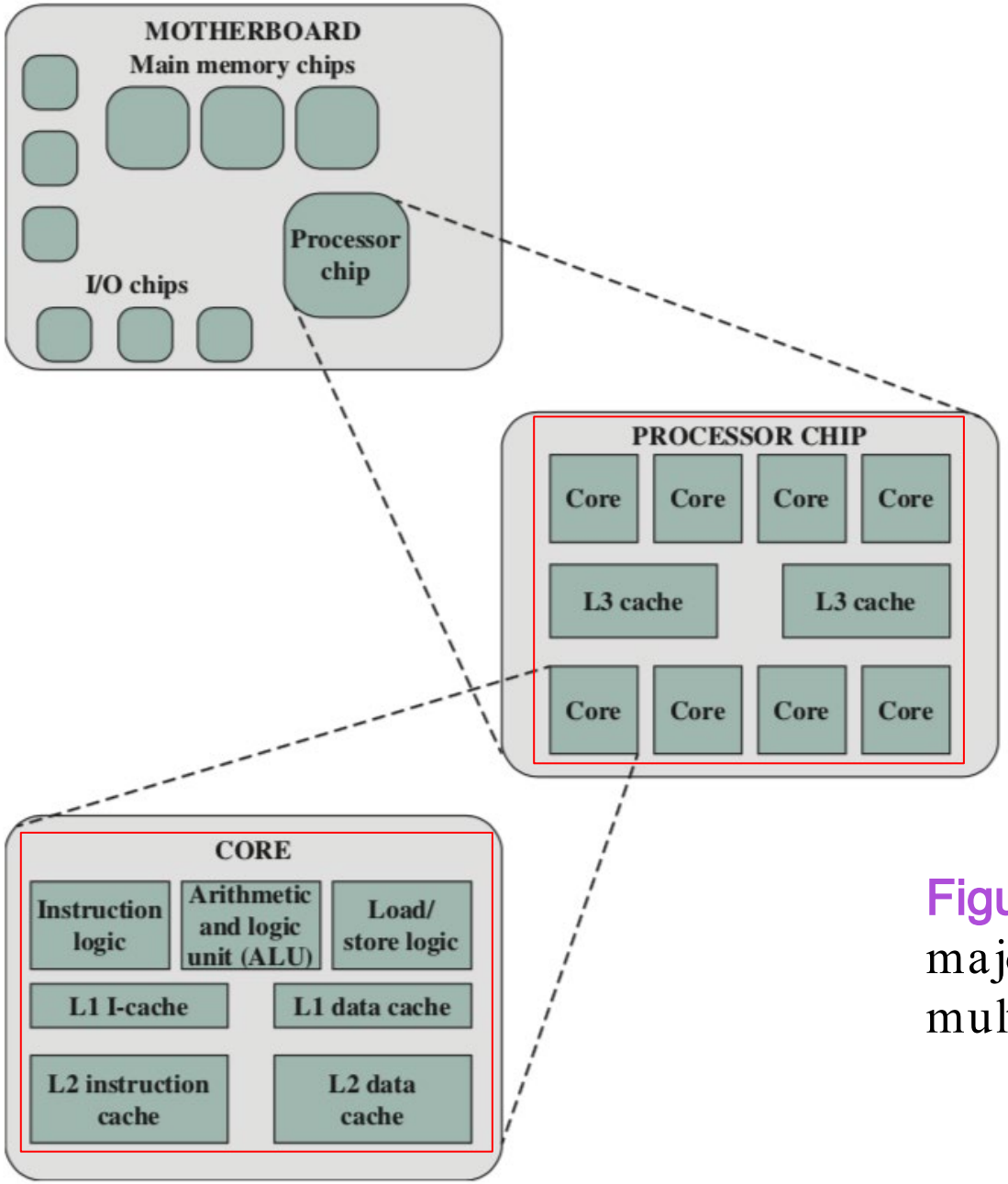


Figure : Simplified view of major elements for a multicore computer

Functions

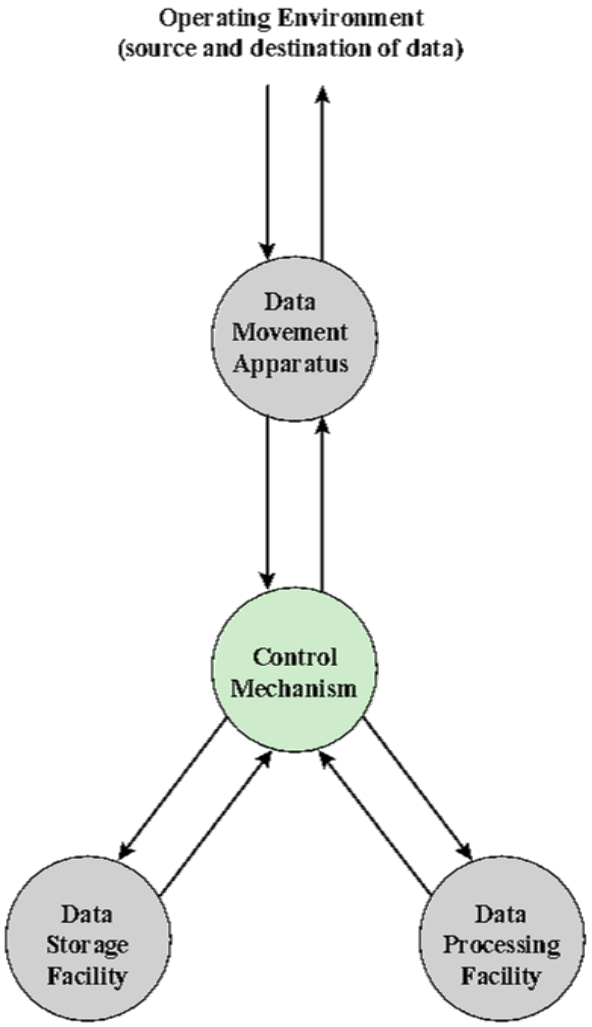


Figure : A functional view of the computer.

- Both the structure and functioning of a computer are simple.
- Figure depicts the basic functions that a computer can perform.
- In general terms, there are only four:
  - Data processing
  - Data storage
  - Data movement
  - Control

Table: The task for each function component.

Functions	Task
Data Processing	Process data in variety of forms and requirements.
Data Storage	Short and long term data storage for retrieval and update.
Data Movement	Moves data between the computer and outside world.
Control	Control of process, move and store data using instruction.

Functions are performed through ***programs***

Computer Program

1

- A sequence of steps.
- For each step, a computer function is executed.
- For each operation:
  - a different/new set of control signals is needed.
  - a unique code (instruction) is provided. *e.g.* `ADD`, `MOVE`
- A hardware segment accepts the code and issues the control signals.

Program: Execution

1

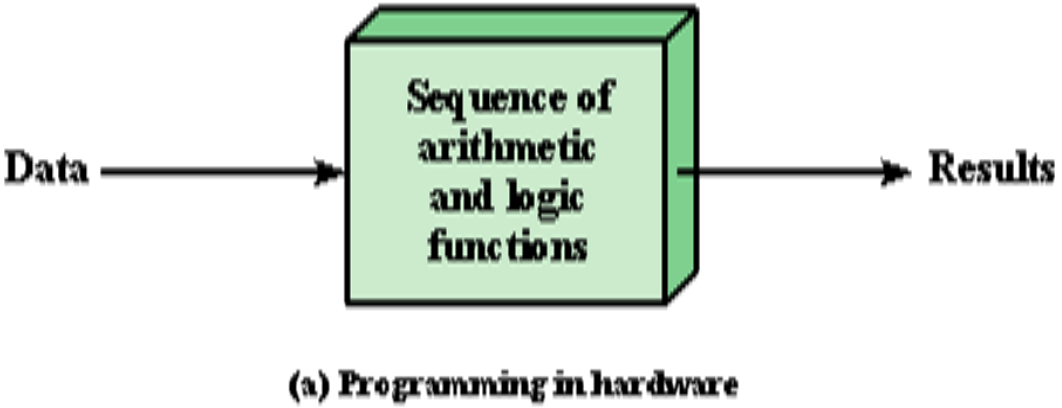
- Two approaches:

Approach 1: Hardwired

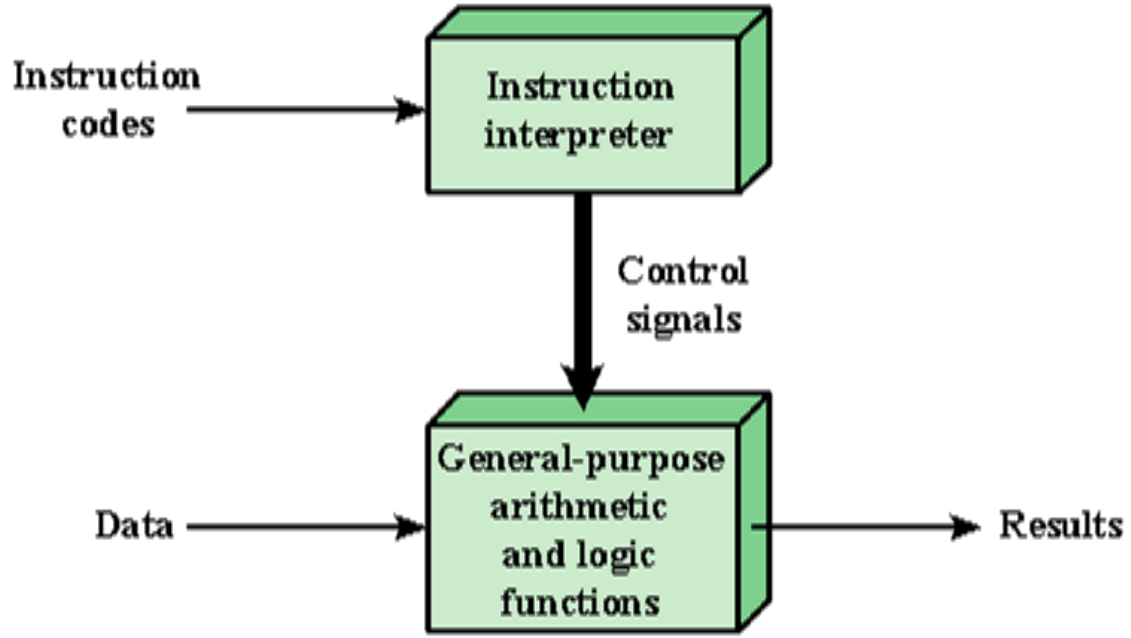
- Built into (direct wired into) computers hardware.
- Uses a sequence of arithmetic and logic functions
- Provides high speed but inflexible to change.

Approach 2: Software

- Control signals through instruction codes.
- Needs an interpreter to “speak to the machine”.
- While less speed, it is easily reprogrammable.



(a) Programming in hardware



(b) Programming in software

Figure : Hardware and software approaches.

Example of 'hardwired approach' (*embedded systems*)

1

- The use of electronics and software within a product
- Billions of computer systems are produced each year that are embedded within larger devices
- Today many devices that use electric power have an embedded computing system













Example of 'software approach' (*general purpose CPU*)

1

- The use of general purpose CPU within a computing device product
- we have PC, laptop, smartphone, mobile tablet, gaming console, … and lot of







Module 1
Introduction

1.1 An Overview

1.2 Components of Computer

1.3 Computer Structure and Functions

1.4 Computer Evolution

1.5 Computer Level Hierarchy

1.6 An Example System

1.7 Summary

- Historical Development

1.4 Computer Evolution

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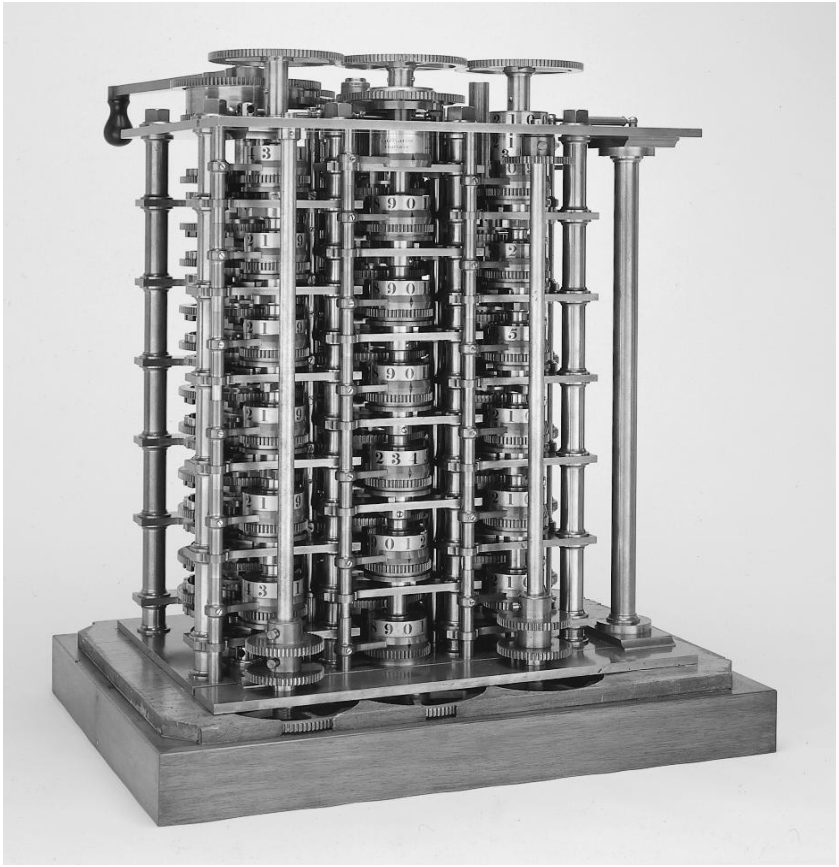
Historical Development

- The evolution of computers can be divided into generations.
- Each generation being defined by the technology used to build the machine.

Gen.	Years	Technology	Typical Speed (Operation / sec)
0	1642 – 1946	Mechanical Calculating Machines	< 40,000
1st	1946 – 1957	Vacuum Tube Computers	40,000
2nd	1957 – 1964	Transistorized Computers	200,000
3rd 4th	1965 – 1978	Integrated Circuit Computers	1,000,000
5th	1978 – 1991	Very large scale integration (VLSI)	100,000,000
6th	1991 – ????	Ultra large scale integration (ULSI)	> 1,000,000,000

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.12.
William Stallings (2016). *Computer Organization and Architecture: Designing for Performance* (10th Edition). United States: Pearson Education Limited, p.17.

Generation 0: Mechanical Calculator



The Babbage Difference Engine (1832)

25,000 parts

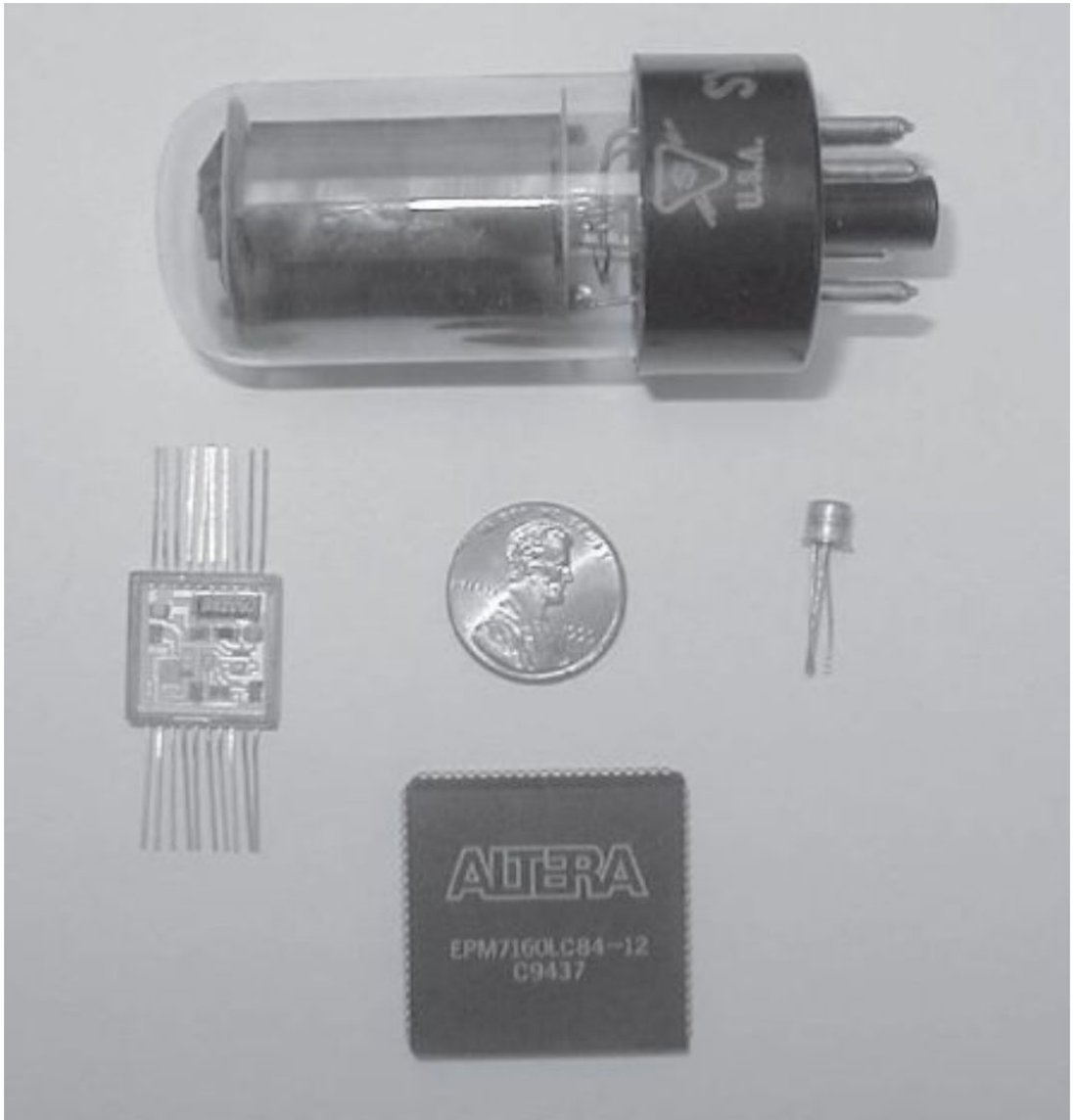
cost: £17,470

Charles Babbage (1791-1871)



Construction of a machine called "Difference Engine."

Comparison of Computer Technologies

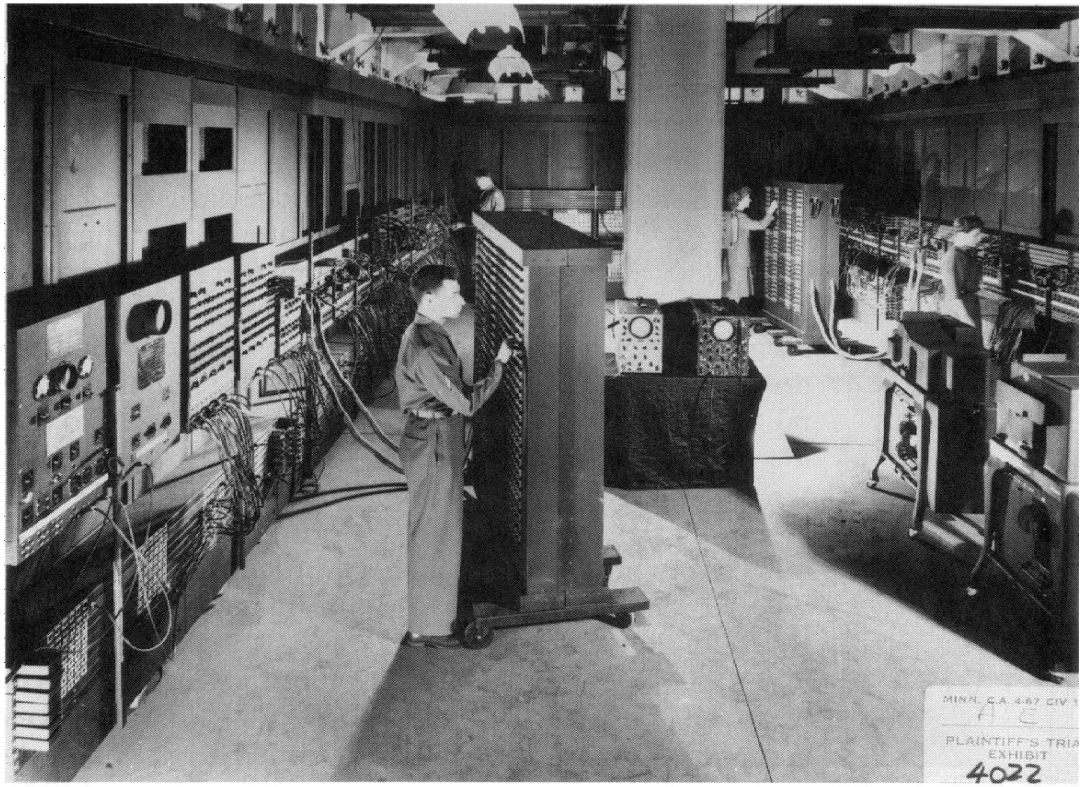


- Clockwise, starting from the top:
  - 1) Vacuum tube
  - 2) Transistor
  - 3) Chip containing 3200 2-input NAND gates (VLSI)
  - 4) Integrated circuit package (the small silver square in the lower left-hand corner is an integrated circuit)

Courtesy of Linda Null.

1st Generation: Vacuum Tubes Computers

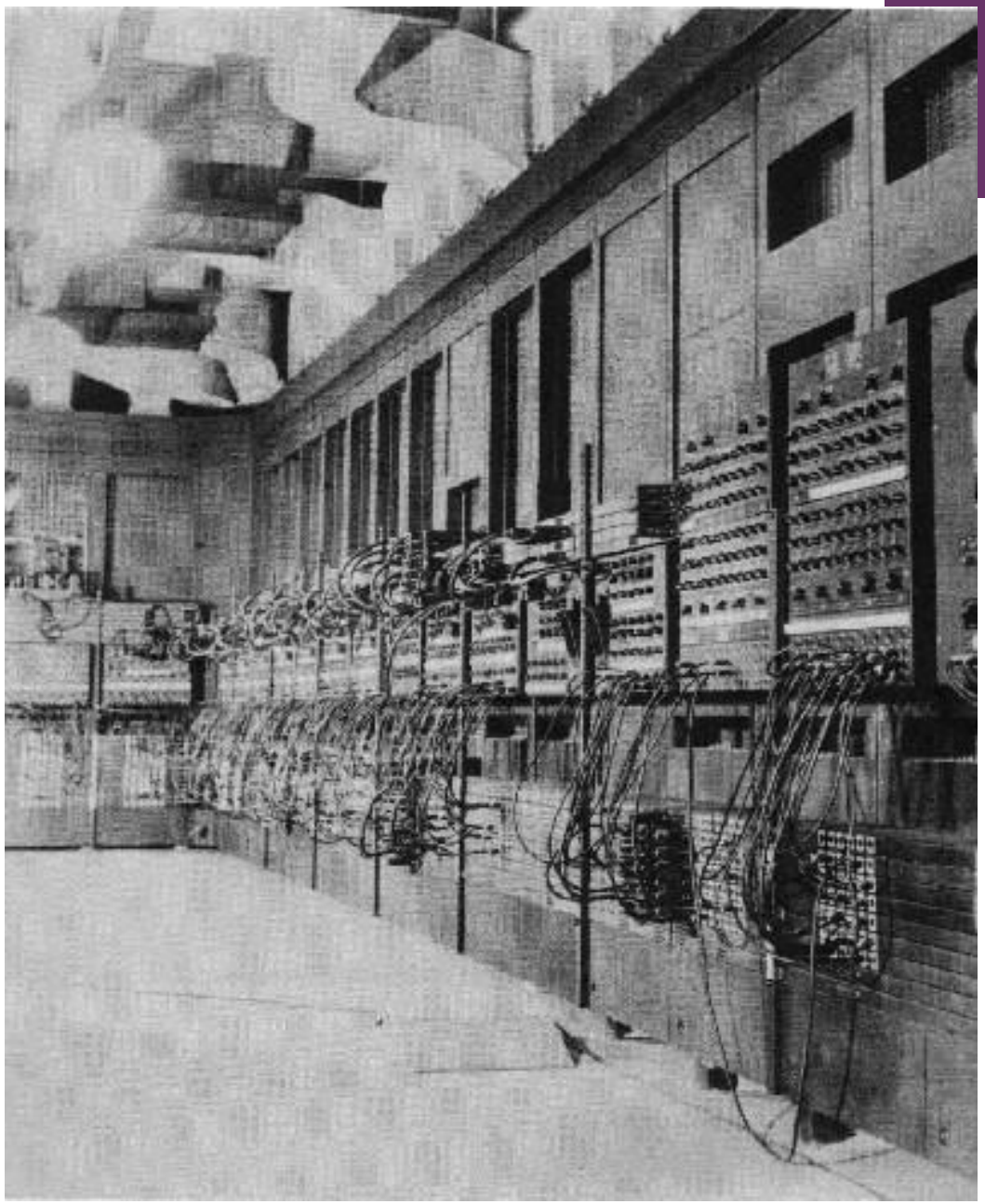
- ENIAC
  - Electronic Numerical Integrator And Computer
  - Designed and constructed at the University of Pennsylvania
  - Started in 1943 – completed in 1946
  - By John Mauchly and John Eckert



William Stallings (2013). *Computer Organization and Architecture: Designing for Performance* (9th Edition). United States: Pearson Education Limited, p.39.
Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.27-30.
https://pbs.twimg.com/media/DcHPdYZV4AAPZb3.jpg

ENIAC - (1946)

- World's first general purpose electronic digital computer
- Army's Ballistics Research Laboratory (BRL) needed a way to supply trajectory tables for new weapons accurately and within a reasonable time frame
- Was not finished in time to be used in the war effort

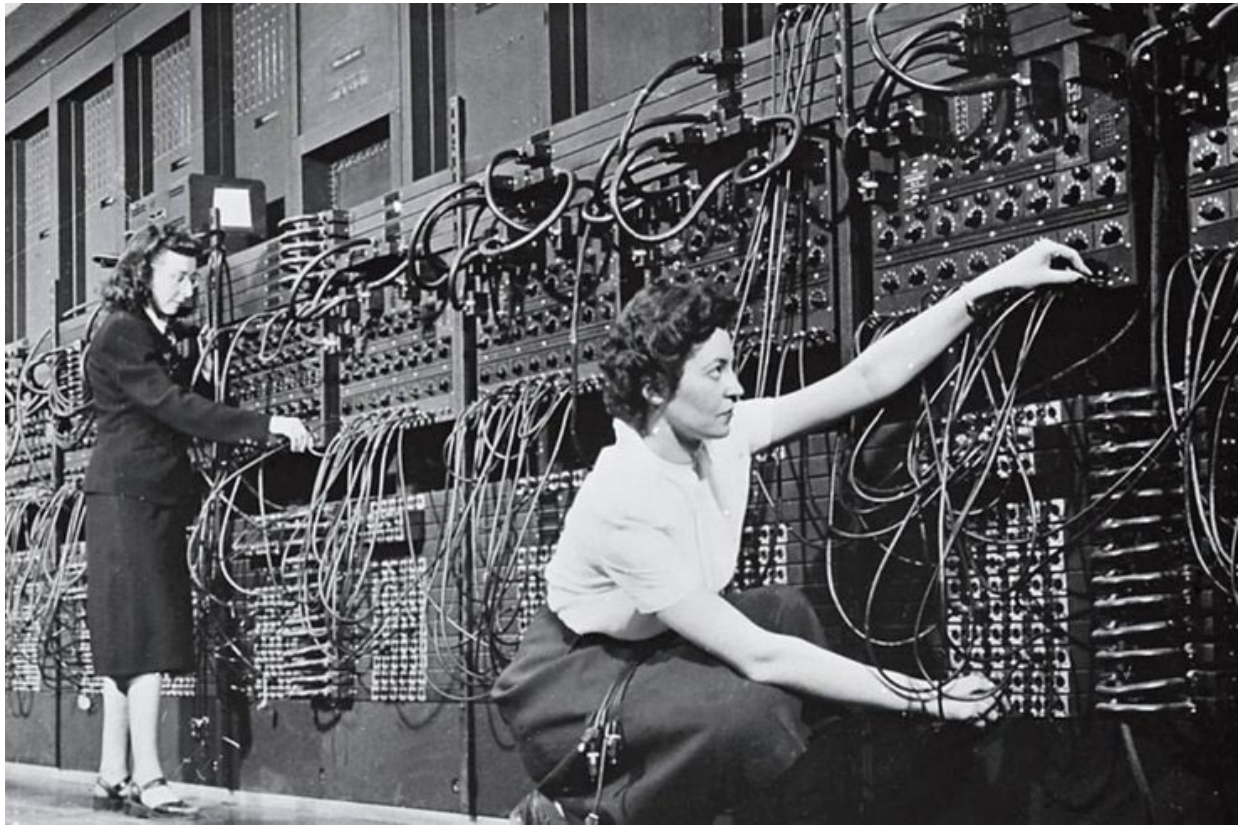


ENIAC – Key Features …

- Weighed 30 tons
- Occupied 1500 square feet of floor space
- Contained more than 18,000 vacuum tubes
- 140 kW Power consumption
- Capable of 5000 additions per second
- Decimal rather than binary machine
- Memory consisted of 20 accumulators, each capable of holding a 10 digit number
- Major drawback was the need for manual programming by setting switches and plugging/ unplugging cables

1

- In the earliest electronic computing machines, programming was synonymous with connecting wires to plugs.



William Stallings (2013). *Computer Organization and Architecture: Designing for Performance* (9th Edition). United States: Pearson Education Limited, p.39.
Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.27-30.
https://pbs.twimg.com/media/DcHPdYZV4AAPZb3.jpg

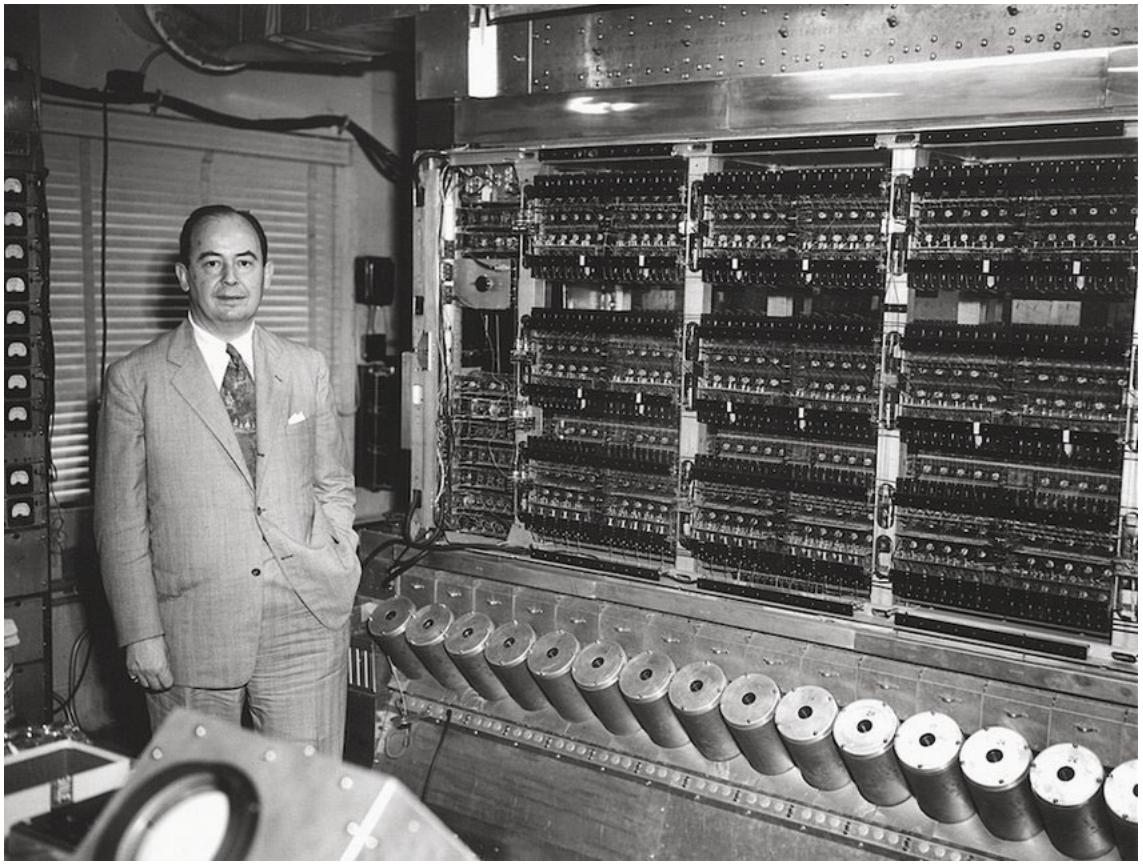
1st Generation:

The von Neumann Model

1

(pronounced *von noy-man*)

- All stored-program computers have come to be known as *von Neumann architecture* (or *model*)
- IAS computer
  - Princeton Institute for Advanced Studies
  - Prototype of all subsequent general-purpose computers
  - Completed in 1952



William Stallings (2013). *Computer Organization and Architecture: Designing for Performance* (9th Edition). United States: Pearson Education Limited, p.39.
Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.27-30.

- This von Neumann model used for designing and building the first generation (Vacuum Tubes Computer).

- The model based on the following three characteristics :

1. The computer consists of four main sub-systems:
   (Memory, ALU, Control Unit, I/O)
2. Program is stored in memory during execution.
3. Program instructions are executed sequentially.

- Single path between the main memory and the CPU, forcing alternation of instruction and execution cycles.
- This is often referred to as the:

von Neumann bottleneck [2]

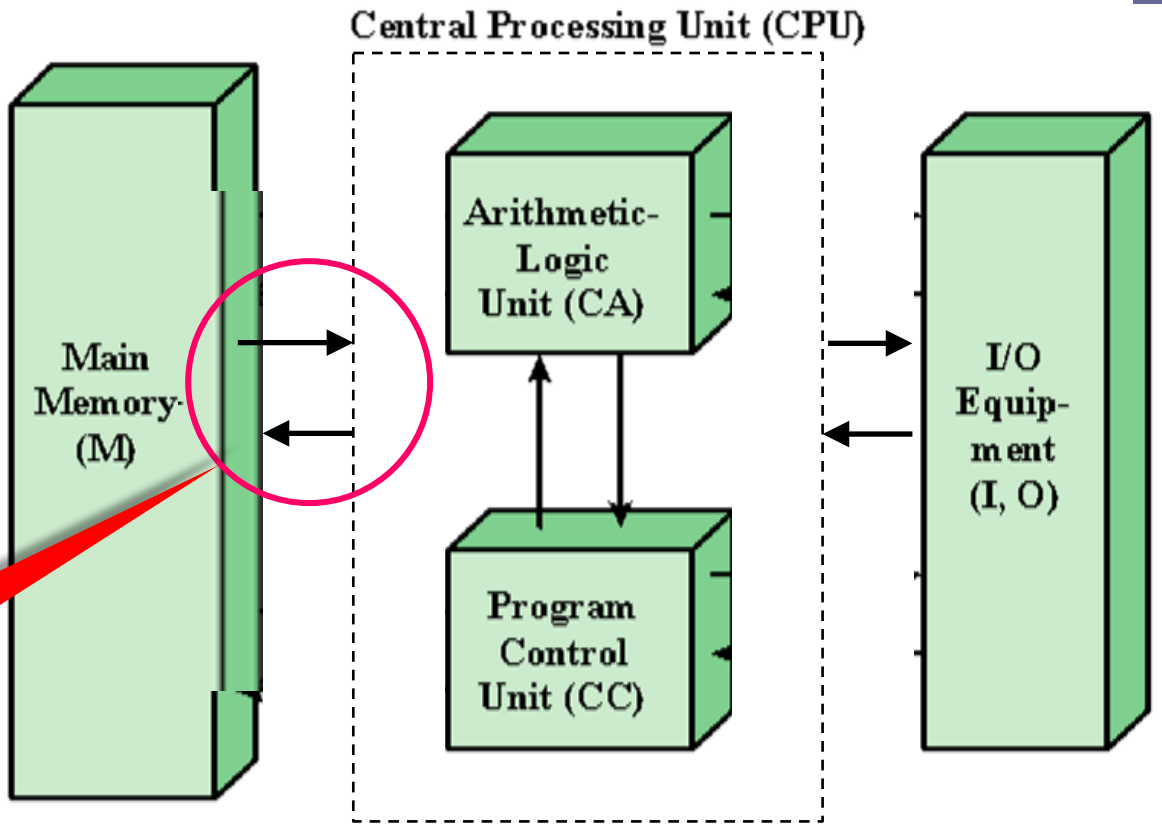


Figure : Structure of von Neumann machine.

1 William Stallings (2013). *Computer Organization and Architecture: Designing for Performance* (9th Edition). United States: Pearson Education Limited, p.40.
2 Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.28.



- Conventional stored-program computers have undergone many incremental improvements over the years.

- These improvements include by adding specialized buses, floating-point units, and cache memories, to name only a few.

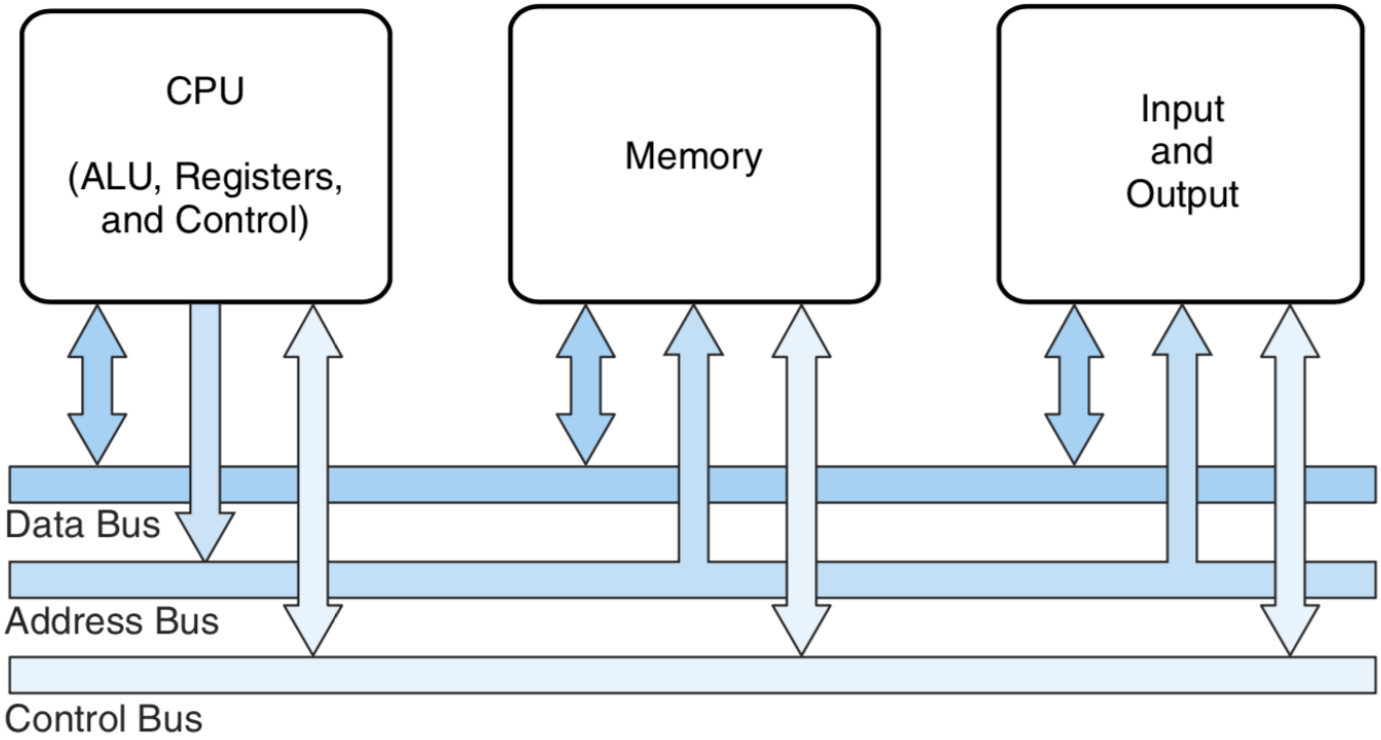


Figure : The modified von Neumann architecture, adding a system bus.

- Departure from the classic von Neumann architecture
  - hence the non von Neumann model.

- Adding processors and parallel processing are examples of approaches in a non von Neumann model.
  - Adding processors can increase computational throughput.
  - Parallel computers improve speed by doing multiple tasks at one time.

Processors:

- In the late 1960s, high-performance computer systems were equipped with dual processors to increase computational throughput.
- In the 1970s supercomputer systems were introduced with 32 processors.
- Supercomputers with 1,000 processors were built in the 1980s.
- In 1999, IBM announced its *Blue Gene* system containing over 1 million processors.



https://upload.wikimedia.org/wikipedia/commons/d/d3/IBM_Blue_Gene_P_supercomputer.jpg

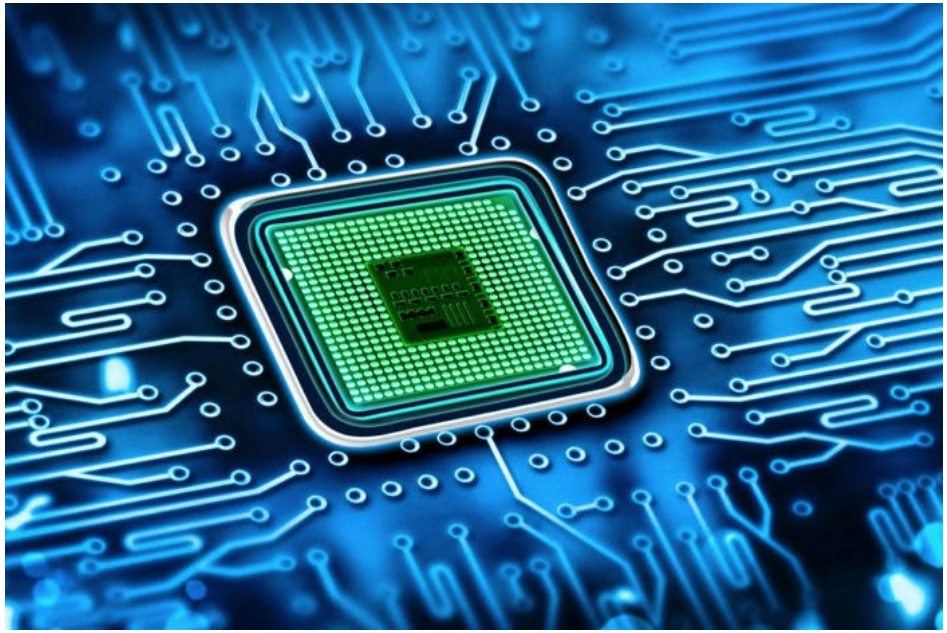
Parallel Processing:

- Parallel processing is currently the most popular and the only one method of providing increased computational power.
- More radical systems have reinvented the fundamental concepts of computation.
- These advanced systems include genetic computers, quantum computers, and dataflow systems.

- At this point, it is unclear whether any of these systems will provide the basis for the next generation of computers.

4th Generation: Microprocessors

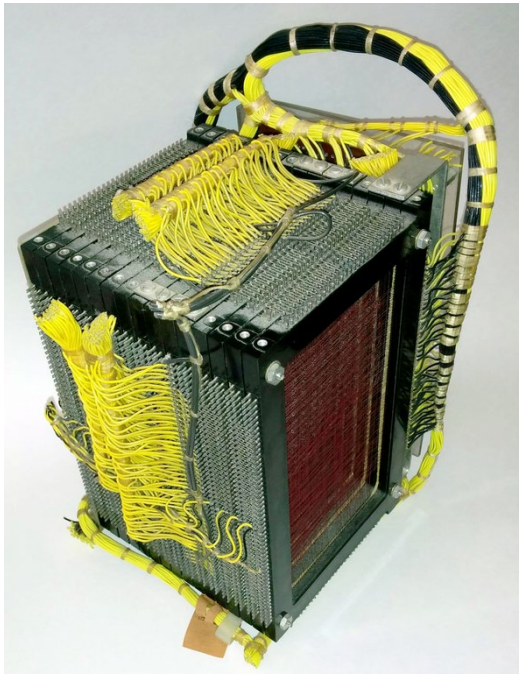
- Microprocessor : all CPU components on a single chip (i.e. integrated circuit technology)

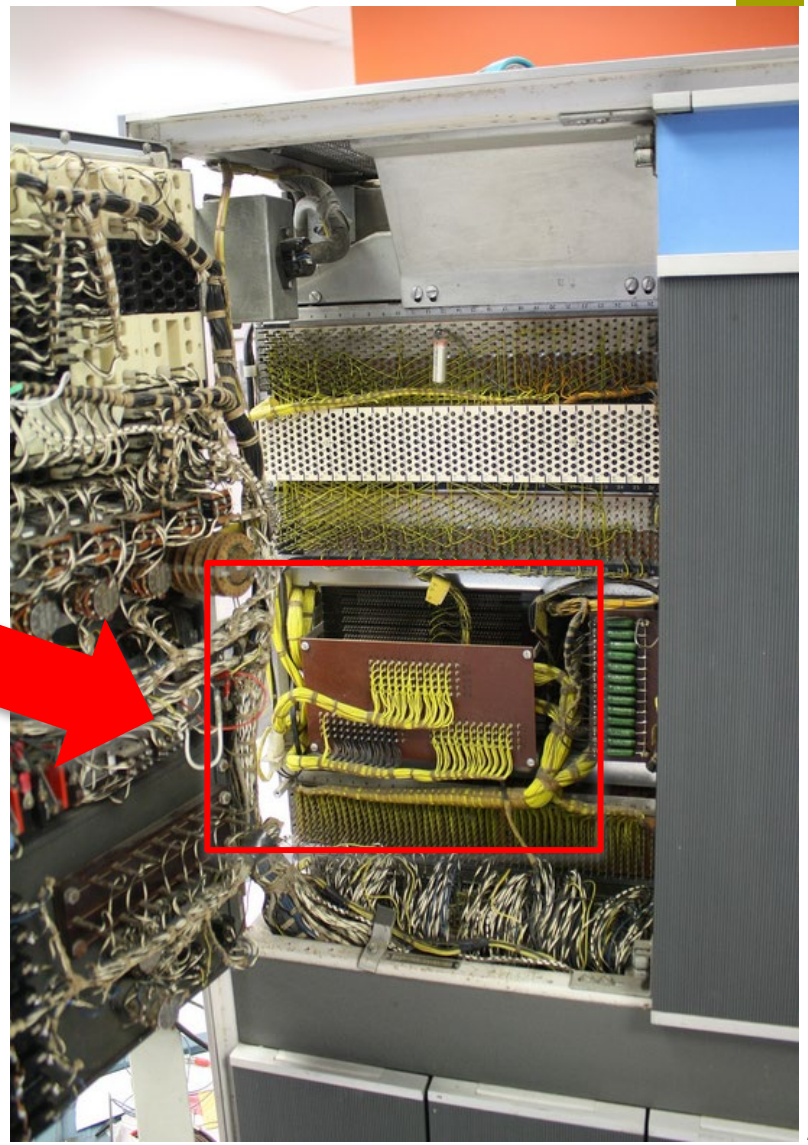


- Semiconductor memory : used of integrated circuit chips to construct computer memory

Core Memory – IBM Mainframe (1960)

1





Fairchild Memory (1970) – 256-bit bipolar SRAM



Modern RAMs

DDR



DDR2



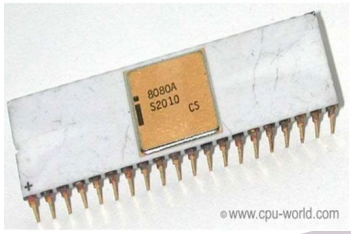
DDR3



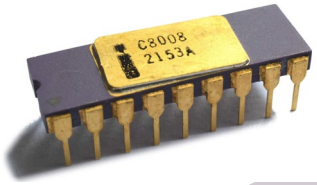
DDR4



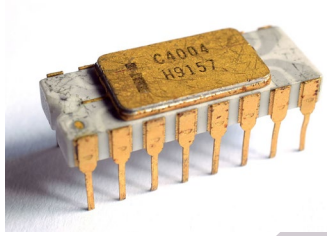
Integrated Circuits - based



Intel 8080



Intel 8008



Intel 4004

1974

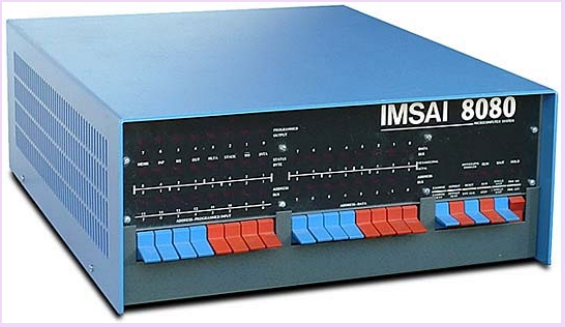
- Intel's first general purpose microprocessor
- Designed to be the CPU of a general purpose microcomputer.

1972

- 8 bits
- Both designed for specific applications.

1971

- First microprocessor
- 4 bits



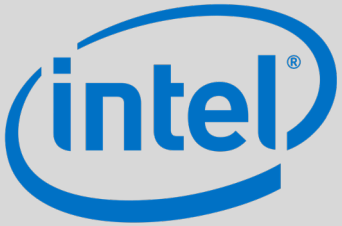
Used in first personal computer – Altair

Figure: Early Evolution of Intel Processors

The Evolution of the Intel x86 Architecture

1

- Intel has ranked as the number 1 maker of microprocessors for non-embedded systems for decades.



VLSI - based

- 8086 (1978)
  - much more powerful.
  - 16 bits
  - instruction cache, pre-fetch few instructions.
  - 8088 (8 bit external bus) used in first IBM PC.
- 80286 (1982)
  - 16 MB memory addressable.
- 80386 (1985)
  - First 32 bit design.
  - Support for multitasking that run multiple programs at the same time.

1

- 80486 (1989)
  - sophisticated powerful cache and instruction pipelining.
  - built in maths co-processor.

ULSI - based

- Pentium (1993)
  - Superscalar technique - multiple instructions executed in parallel.



- Pentium Pro (1995)
  - 64-bit computing
  - increased superscalar organization.
  - branch prediction.
  - data flow analysis.
  - speculative execution.



- Pentium II (1997)
  - MMX technology
  - graphics, video and audio processing.
- Pentium III (1999)
  - Additional floating point instructions for 3D graphics.
- Pentium 4 (2000)
  - Further floating point and multimedia enhancements.



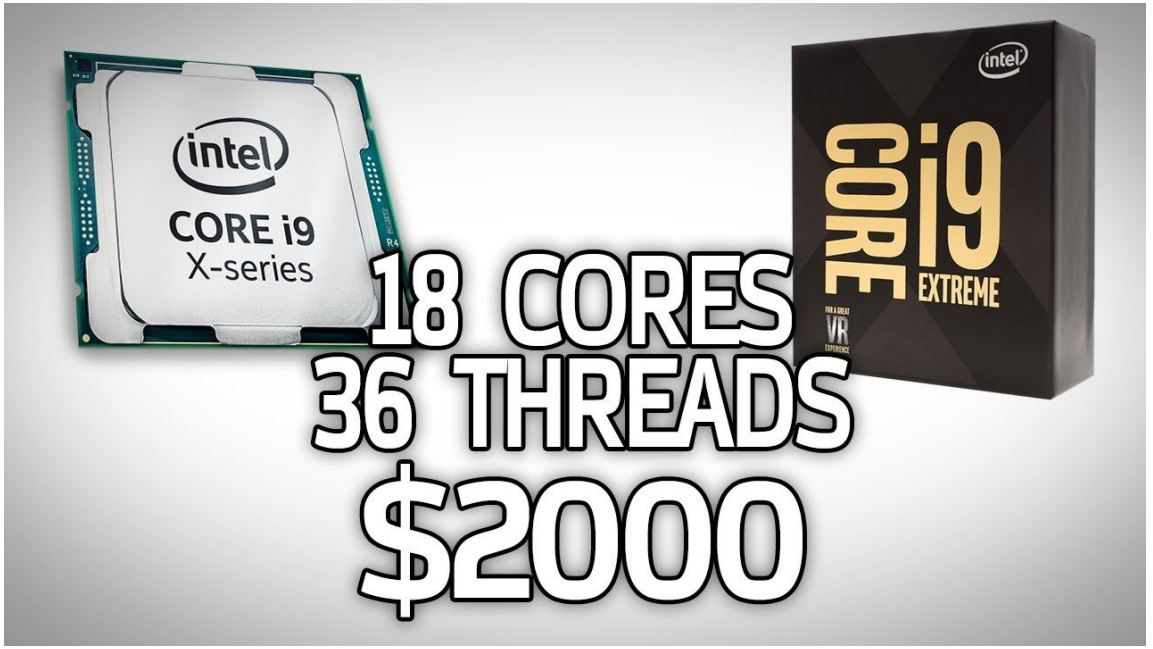
- Itanium (2001)
  - 64 bit.



- Core Duo (2006)
  - start of multicore processors.

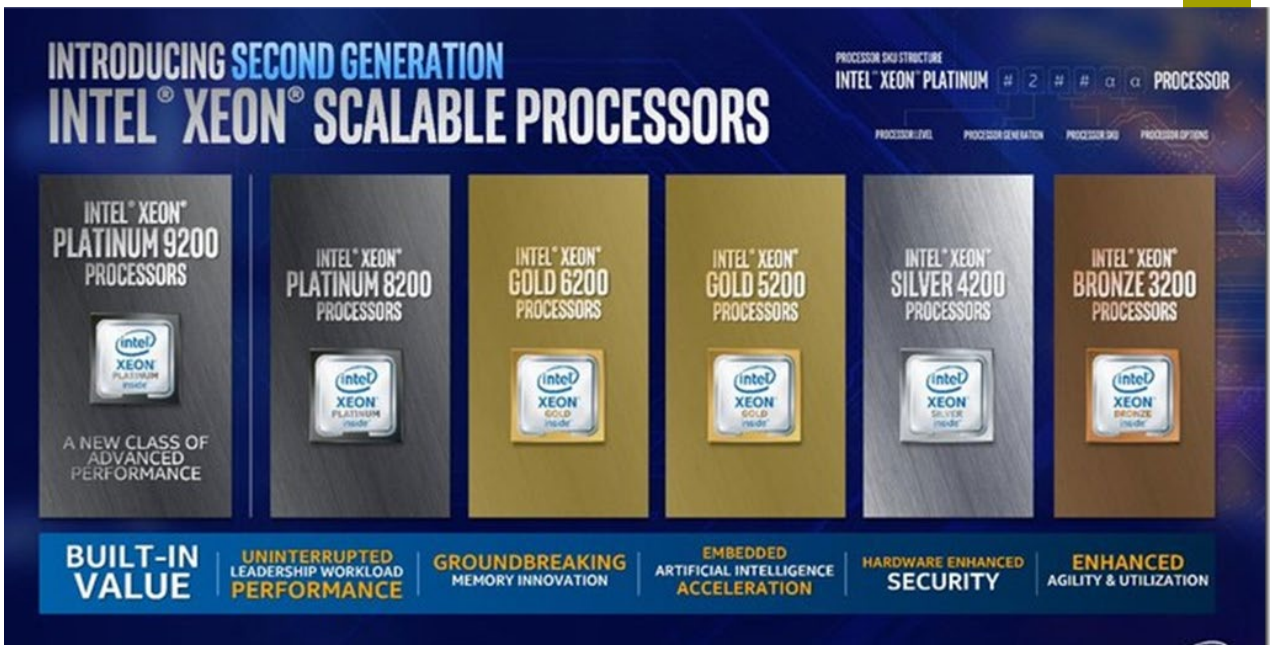


- ... Latest Core (as of Feb, 2020)
  - i.e. Core i9 offerings up to 18 processors per chip



1

- Xeon Processor
  - Very high grade CPU by Intel (i.e. professional / enterprise grade of processors)



  - Demanding heavy-duty tasks that need constant high processing power.
  - processors for artificial intelligence, servers, 3D modelling and rendering, and content creation.

Guess the prices for top notch Xeon CPUs ???



	Multiprocessors (8-way)										
8253	Xeon Platinum	$ 3,115.00	2 April 2019	16	32	2.2 GHz	3 GHz	125 W	22 MiB	DDR4-2933	1 TiB
8256	Xeon Platinum	$ 7,007.00, $ 7,014.00	2 April 2019	4	8	3.8 GHz	3.9 GHz	105 W	16.5 MiB	DDR4-2933	1 TiB
8260	Xeon Platinum	$ 4,702.00	2 April 2019	24	48	2.4 GHz	3.9 GHz	165 W	35.75 MiB	DDR4-2933	1 TiB
8260L	Xeon Platinum	$ 12,599.00	2 April 2019	24	48	2.4 GHz	3.9 GHz	165 W	35.75 MiB	DDR4-2933	4.5 TiB
8260M	Xeon Platinum	$ 7,705.00	2 April 2019	24	48	2.4 GHz	3.9 GHz	165 W	35.75 MiB	DDR4-2933	2 TiB
8260Y	Xeon Platinum	$ 5,320.00	2 April 2019	24	48	2.4 GHz	3.9 GHz	165 W	35.75 MiB	DDR4-2933	1 TiB
8268	Xeon Platinum	$ 6,302.00	2 April 2019	24	48	2.9 GHz	3.9 GHz	205 W	35.75 MiB	DDR4-2933	1 TiB
8270	Xeon Platinum	$ 7,405.00	2 April 2019	26	52	2.7 GHz	4 GHz	205 W	35.75 MiB	DDR4-2933	1 TiB
8276	Xeon Platinum	$ 8,719.00	2 April 2019	28	56	2.2 GHz	4 GHz	165 W	38.5 MiB	DDR4-2933	1 TiB
8276L	Xeon Platinum	$ 16,616.00	2 April 2019	28	56	2.2 GHz	4 GHz	165 W	38.5 MiB	DDR4-2933	4.5 TiB
8276M	Xeon Platinum	$ 11,722.00	2 April 2019	28	56	2.2 GHz	4 GHz	165 W	38.5 MiB	DDR4-2933	2 TiB
8280	Xeon Platinum	$ 10,009.00	2 April 2019	28	56	2.7 GHz	4 GHz	205 W	38.5 MiB	DDR4-2933	1 TiB
8280L	Xeon Platinum	$ 17,906.00	2 April 2019	28	56	2.7 GHz	4 GHz	205 W	38.5 MiB	DDR4-2933	4.5 TiB
8280M	Xeon Platinum	$ 13,012.00	2 April 2019	28	56	2.7 GHz	4 GHz	205 W	38.5 MiB	DDR4-2933	2 TiB
8284	Xeon Platinum	$ 15,460.00	10 July 2019	28	56	3 GHz	4 GHz	240 W	38.5 MiB	DDR4-2933	1 TiB

no of cores

max RAM

Table: (a) 1990s processors (b) Recent processors.

(a)

	486TM SX	Pentium	Pentium Pro	Pentium II
Introduced	1991	1993	1995	1997
Clock speeds	16–33 MHz	60–166 MHz,	150–200 MHz	200–300 MHz
Bus width	32 bits	32 bits	64 bits	64 bits
Number of transistors	1.185 million	3.1 million	5.5 million	7.5 million
Feature size (μm)	1	0.8	0.6	0.35
Addressable memory	4 GB	4 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	8 kB	8 kB	512 kB L1 and 1 MB L2	512 kB L2

(b)

	Pentium III	Pentium 4	Core 2 Duo	Core i7 EE 4960X
Introduced	1999	2000	2006	2013
Clock speeds	450–660 MHz	1.3–1.8 GHz	1.06–1.2 GHz	4 GHz
Bus width	64 bits	64 bits	64 bits	64 bits
Number of transistors	9.5 million	42 million	167 million	1.86 billion
Feature size (nm)	250	180	65	22
Addressable memory	64 GB	64 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	512 kB L2	256 kB L2	2 MB L2	1.5 MB L2/15 MB L3
Number of cores	1	1	2	6

L1 (Level 1 – closest to the core)

Module 1
Introduction

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1.3 Computer Structure and Functions
1.4 Computer Evolution
1.5 Computer Level Hierarchy
1.6 An Example System
1.7 Summary

- Overview
- Layers in Virtual Machines

1.5 Computer Level Hierarchy

1

Overview

- Computers consist of many things besides chips.
- Before a computer can do anything worthwhile, it must also use software.
- Writing complex programs requires a "*divide and conquer*" approach, where each program module solves a smaller problem.
- Complex computer systems employ a similar technique through a series of virtual machine layers.

Layers in Virtual Machines

- Each virtual machine layer is an abstraction of the level below it.
- The machines at each level execute their own particular instructions, calling upon machines at lower levels to perform tasks as required.
- Computer circuits ultimately carry out the work.

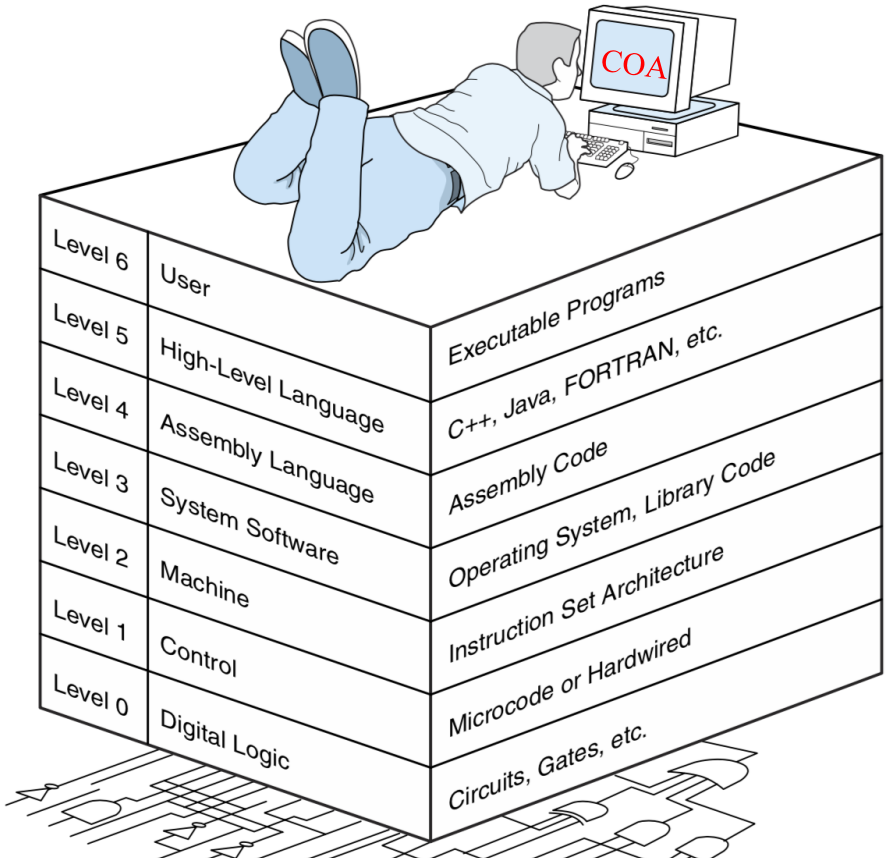
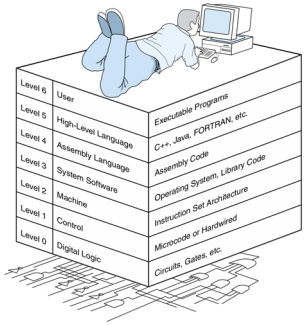


Figure: The Abstract Levels of Modern Computing Systems



1

Table: Commonly accepted layers in virtual machines.

Level 6: The User Level	❑ Program execution and user interface level. ❑ The level with which we are most familiar.
Level 5: High-Level Language Level	❑ The level with which we interact when we write programs in languages such as C, Pascal, Lisp, and Java.
Level 4: Assembly Language Level	❑ Acts upon assembly language produced from Level 5, as well as instructions programmed directly at this level.

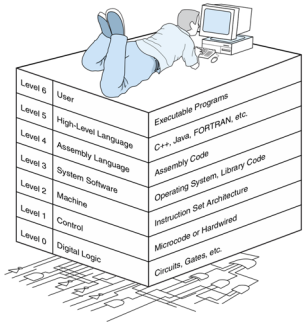
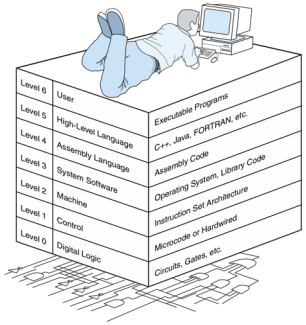


Table: ... Commonly accepted layers in virtual machines.

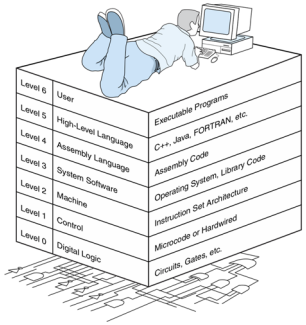
Level 3: System Software Level	❑ Controls executing processes on the system. ❑ Protects system resources. ❑ Assembly language instructions often pass through Level 3 without modification.
Level 2: Machine / ISA Level	❑ Consists of instructions that are particular to the architecture of the machine. ❑ Programs written in machine language need no *compilers*, *interpreters*, or *assemblers*.



1

Table: ... Commonly accepted layers in virtual machines.

Level 1: Control Level	❑ A control unit decodes and executes instructions, and moves data through the system. ❑ Control units can be microprogrammed or hardwired. ➢ A microprogram is a program written in a low-level language that is implemented by the hardware. ➢ Hardwired control units consist of hardware that directly executes machine instructions.



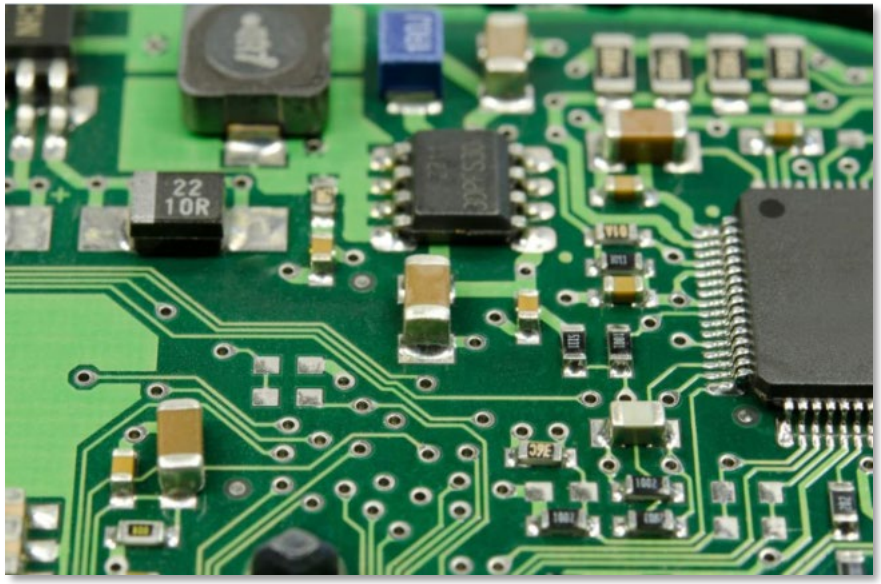


Table: ... Commonly accepted layers in virtual machines.

Level 0: Digital Logic Level	❑ This level is where we find digital circuits (the chips). ❑ Digital circuits consist of gates and wires. ❑ These components implement the mathematical logic & operations of all other levels.

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.27.
http://www.jps-pcb.com/upfile/2016/06/20160629190236_660.png

Module 1
Introduction

1.1 An Overview
1.2 Components of Computer
1.3 Computer Structure and Function
1.4 Computer Evolution
1.5 Computer Level Hierarchy
1.6 An Example System
1.7 Summary

- Overview
- Basic Measurement Terminology

1.6 An Example System

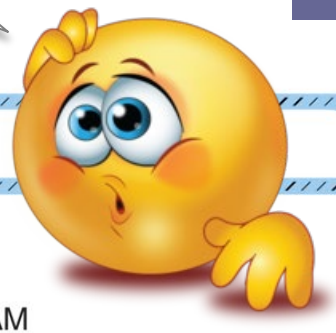
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Overview

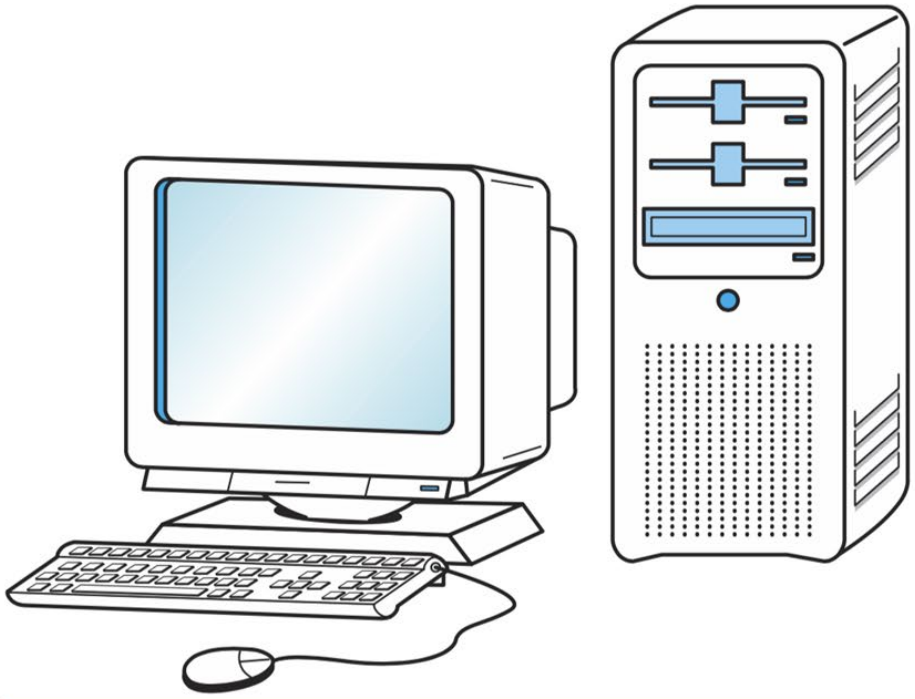
- This section will introduce some of the vocabulary that is specific to computers.
- This jargon can be confusing, imprecise, and intimidating.
- For the sake of discussion, the next slide has provided a facsimile computer advertisement.

What does it all means ?

L1 cache? PCI? MHz?
MB? USB?

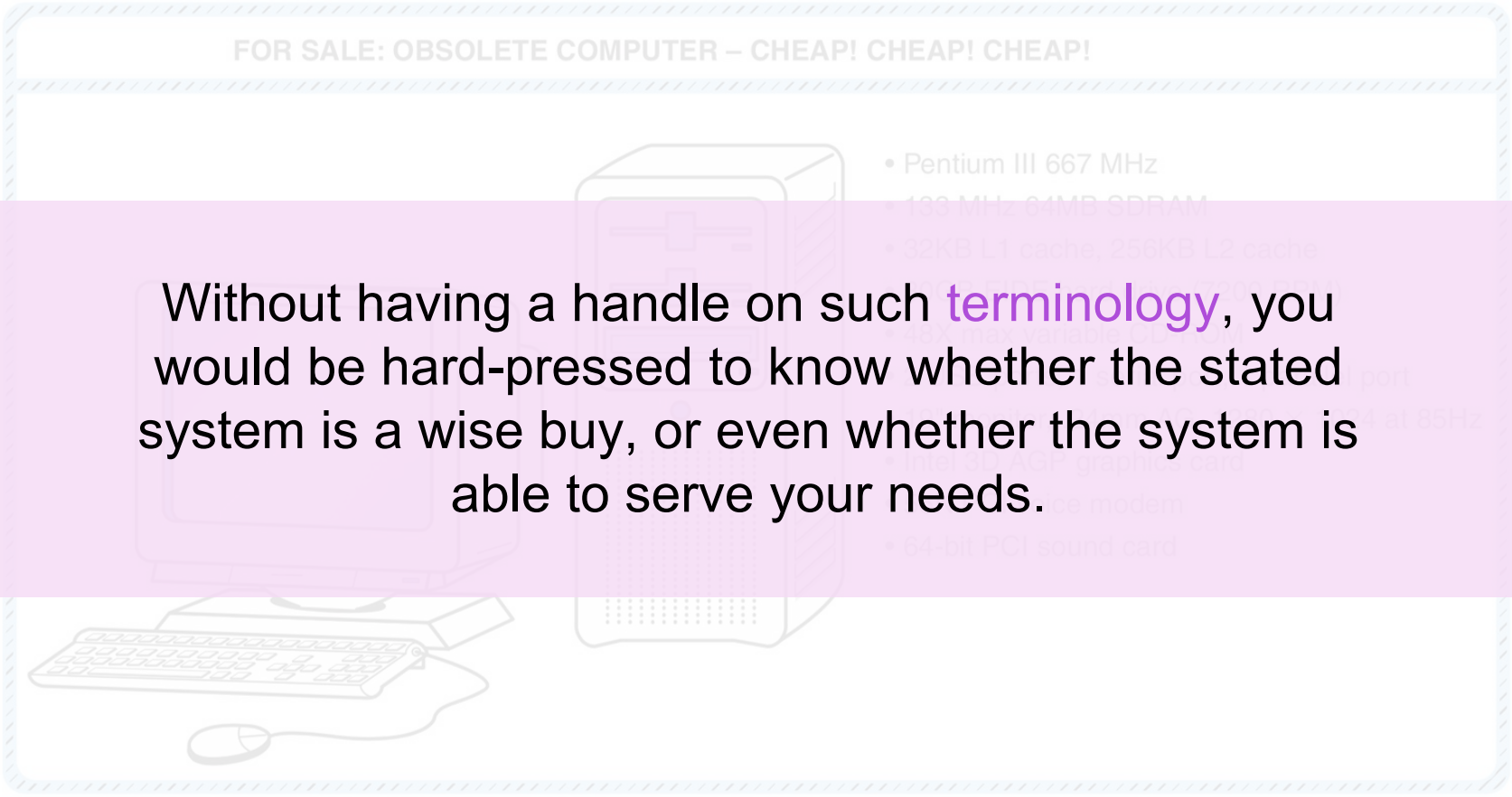


FOR SALE: OBSOLETE COMPUTER – CHEAP! CHEAP! CHEAP!



- Pentium III 667 MHz
- 133 MHz 64MB SDRAM
- 32KB L1 cache, 256KB L2 cache
- 30GB EIDE hard drive (7200 RPM)
- 48X max variable CD-ROM
- 2 USB ports, 1 serial port, 1 parallel port
- 19" monitor, .24mm AG, 1280 × 1024 at 85Hz
- Intel 3D AGP graphics card
- 56K PCI voice modem
- 64-bit PCI sound card

Figure : A typical computer advertisement.



Without having a handle on such terminology, you would be hard-pressed to know whether the stated system is a wise buy, or even whether the system is able to serve your needs.

Figure : A typical computer advertisement.

Basic Measurement Terminology

Kilo- (K)	(1 thousand = $10^3 \approx 2^{10}$)	Milli- (m)	(1 thousandth = $10^{-3} \approx 2^{-10}$)
Mega- (M)	(1 million = $10^6 \approx 2^{20}$)	Micro- (μ)	(1 millionth = $10^{-6} \approx 2^{-20}$)
Giga- (G)	(1 billion = $10^9 \approx 2^{30}$)	Nano- (n)	(1 billionth = $10^{-9} \approx 2^{-30}$)
Tera- (T)	(1 trillion = $10^{12} \approx 2^{40}$)	Pico- (p)	(1 trillionth = $10^{-12} \approx 2^{-40}$)
Peta- (P)	(1 quadrillion = $10^{15} \approx 2^{50}$)	Femto- (f)	(1 quadrillionth = $10^{-15} \approx 2^{-50}$)

(a) Measures of capacity and speed (b) Measures of time and space

Figure: Common Prefixes Associated with Computer Organization and Architecture

- Whether a metric refers to a power of 10 or a power of 2, typically depends upon what is being measured.

Example : Capacity and speed.



- *Hertz* = clock cycles per second (frequency)
  - $1 MHz = 1{,}000{,}000 Hz$
  - Processor speeds are measured in *MHz* or *GHz*.

- *Byte* = a unit of storage
  - $1 KB = 2^{10} = 1024$ *Bytes*.
  - $1 MB = 2^{20} = 1{,}048{,}576$ *Bytes*.
  - Main memory (RAM) is measured in *MB*.
  - Disk storage is measured in *GB* for small systems, *TB* for large systems.

Example: Time and space.



- *Millisecond* = 1 thousandth of a *second*
  - Hard disk drive access times are often 10 to 20 *milliseconds*.
- *Nanosecond* = 1 billionth of a *second*
  - Main memory access times are often 50 to 70 *nanoseconds*.

- *Micron* (micrometer) = 1 millionth of a *meter*
  - Circuits on computer chips are measured in *microns*.

- We note that cycle time is the reciprocal of clock frequency.
- A bus operating at 133*MHz* has a cycle time of 7.52 *nanoseconds*:

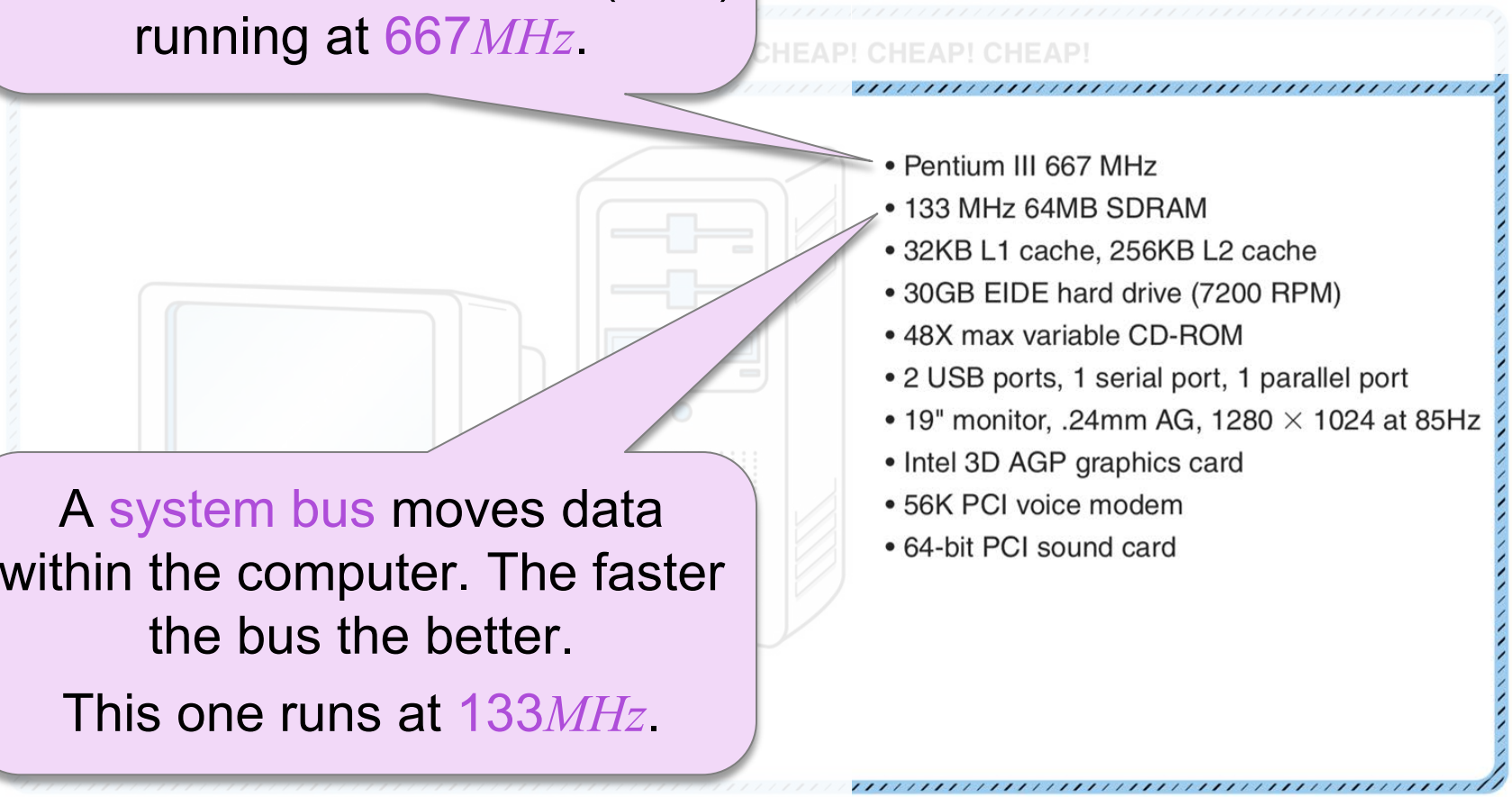
133,000,000 *cycles/second* = 7.52 *ns/cycle*



Lets back to the advertisement …

The microprocessor is the "brain" of the system. It executes program instructions.

This one is a Pentium III (Intel) running at 667*MHz*.



A system bus moves data within the computer. The faster the bus the better.

This one runs at 133*MHz*.

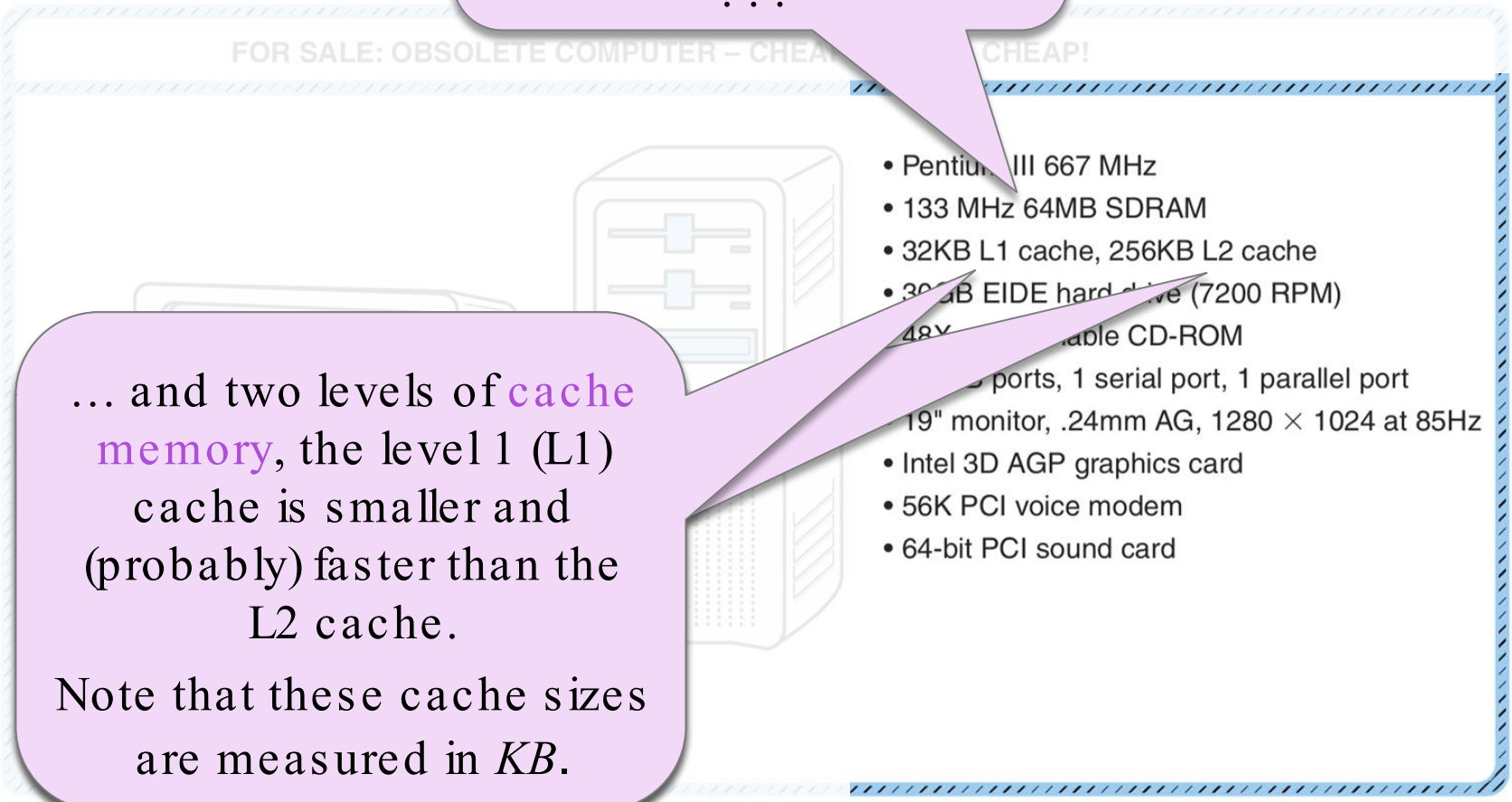
Figure : A typical computer advertisement.



- Computers with large main memory capacity can run larger programs with greater speed than computers having small memories.

- RAM is an acronym for Random Access Memory.
  - Random access means that memory contents can be accessed directly if you know its location.
- Cache is a type of temporary memory that can be accessed faster than RAM.

This system has 64MB of (fast) Synchronous Dynamic RAM (SDRAM) . . .



. . . and two levels of cache memory, the level 1 (L1) cache is smaller and (probably) faster than the L2 cache.

Note that these cache sizes are measured in *KB*.

Figure : A typical computer advertisement.

Hard disk capacity determines the amount of data and size of programs you can store.

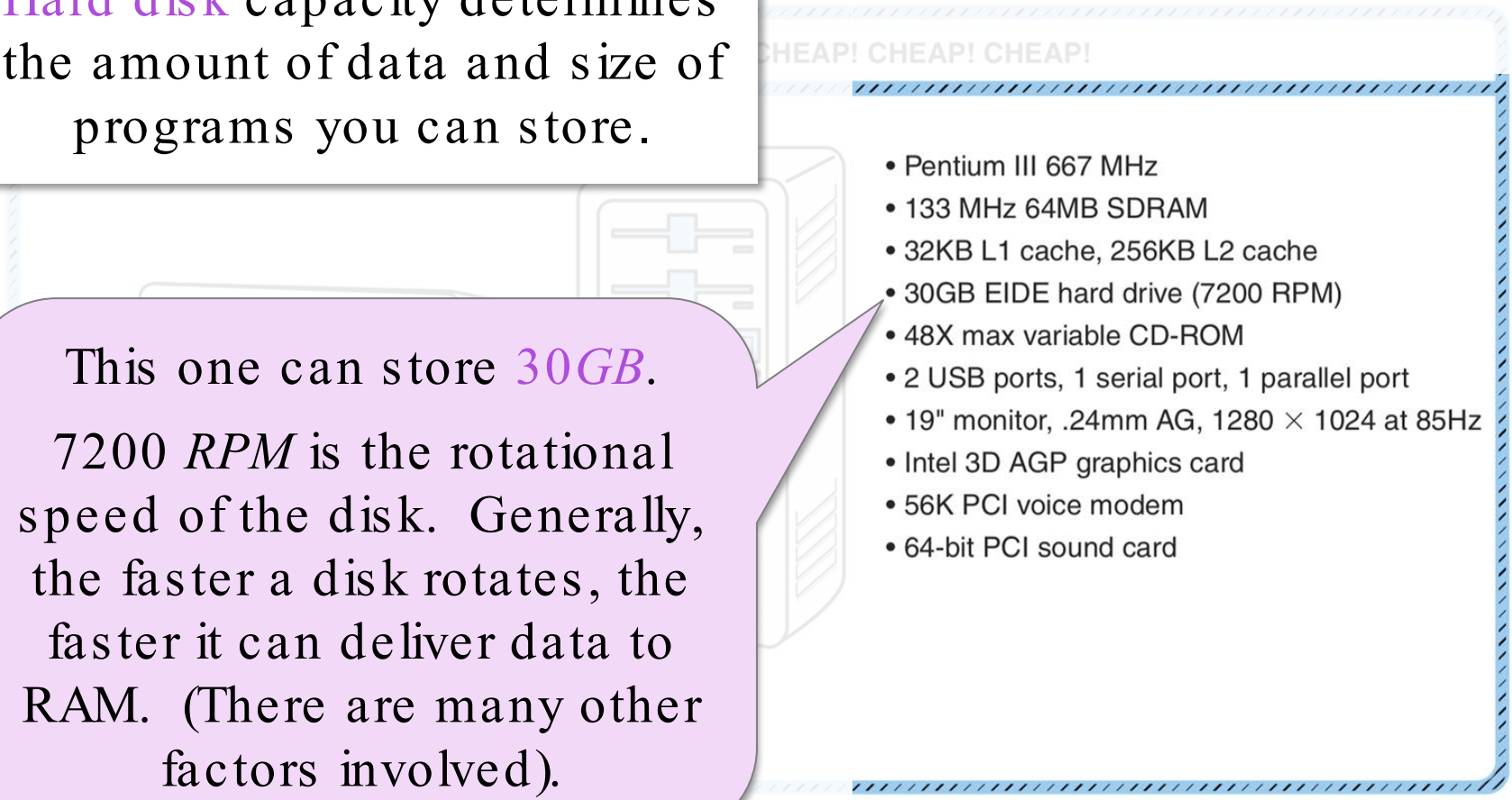


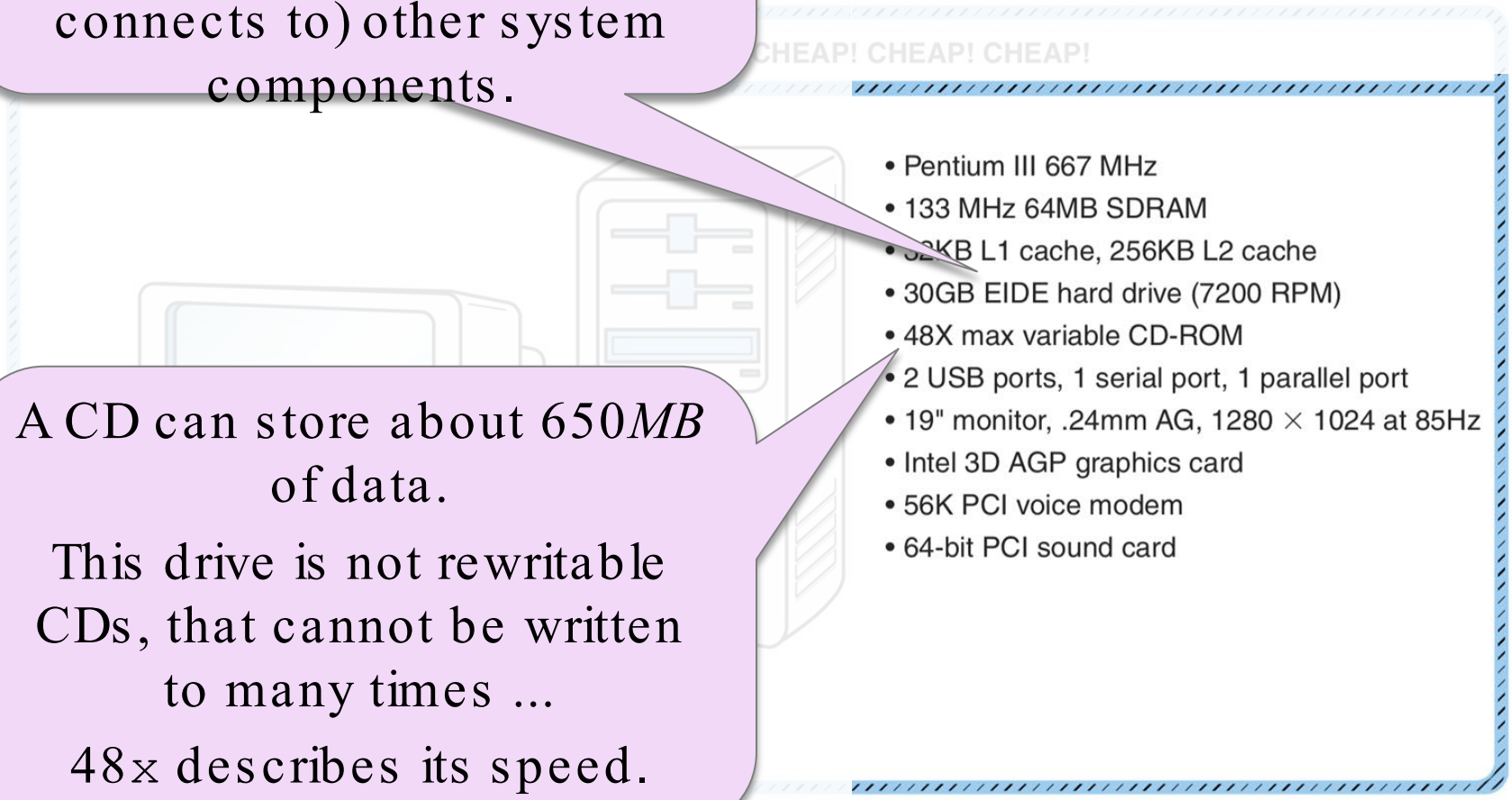
Figure: A typical computer advertisement.

This one can store 30*GB*.

7200 *RPM* is the rotational speed of the disk. Generally, the faster a disk rotates, the faster it can deliver data to RAM. (There are many other factors involved).

1

EIDE stands for *Enhanced Integrated Drive Electronics*, which describes how the hard disk interfaces with (or connects to) other system components.



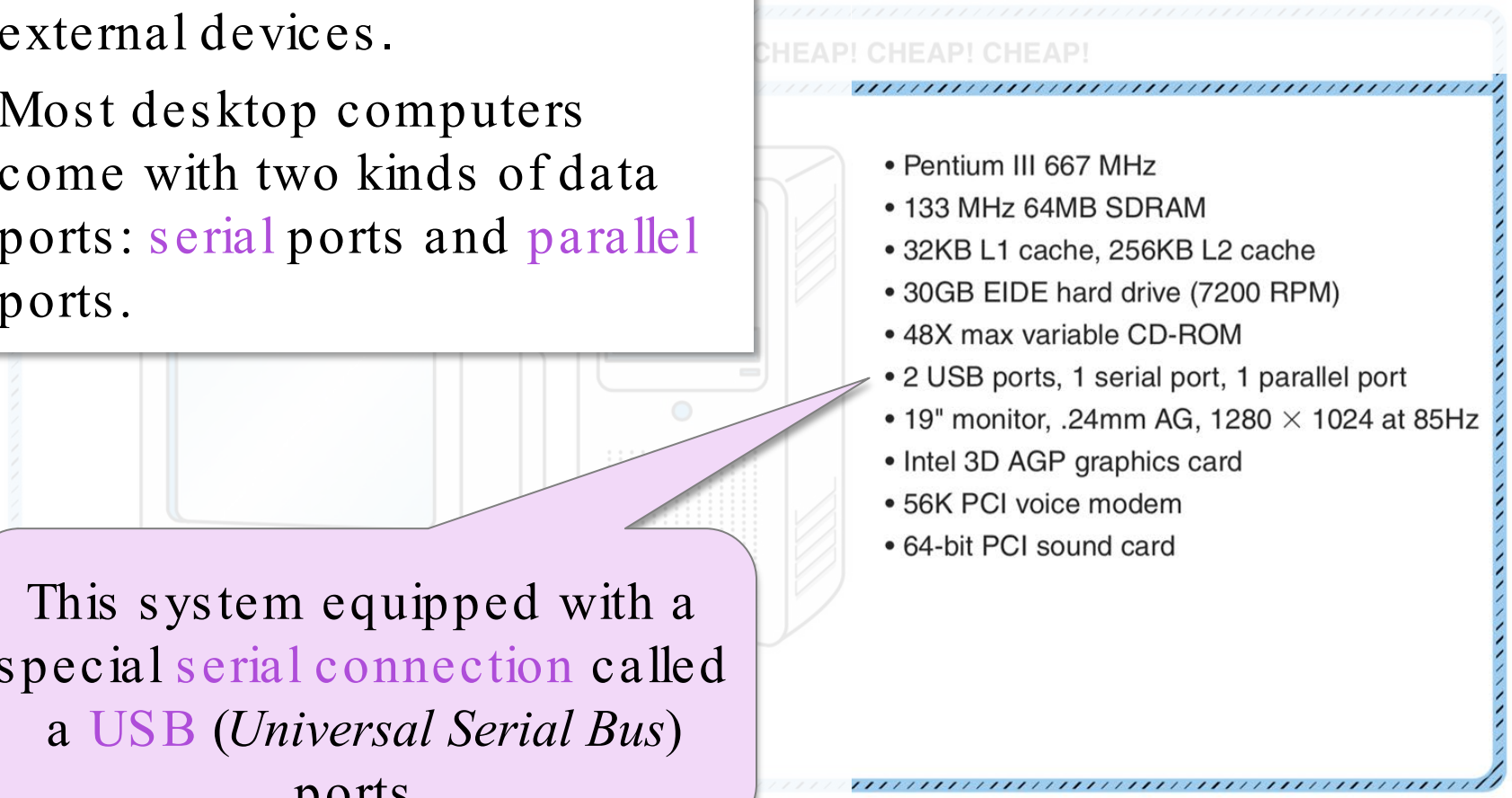
A CD can store about 650*MB* of data.

This drive is not rewritable CDs, that cannot be written to many times ...

48x describes its speed.

Figure: A typical computer advertisement.

- Ports allow movement of data between a system and its external devices.
- Most desktop computers come with two kinds of data ports: serial ports and parallel ports.



This system equipped with a special serial connection called a USB (*Universal Serial Bus*) ports.

Figure: A typical computer advertisement.

- Serial ports send data as a series of pulses along one or two data lines.
- Parallel ports send data as a single pulse along at least eight data lines.
- USB (*Universal Serial Bus*), is an intelligent serial interface that is self-configuring.
  - It supports "plug and play".



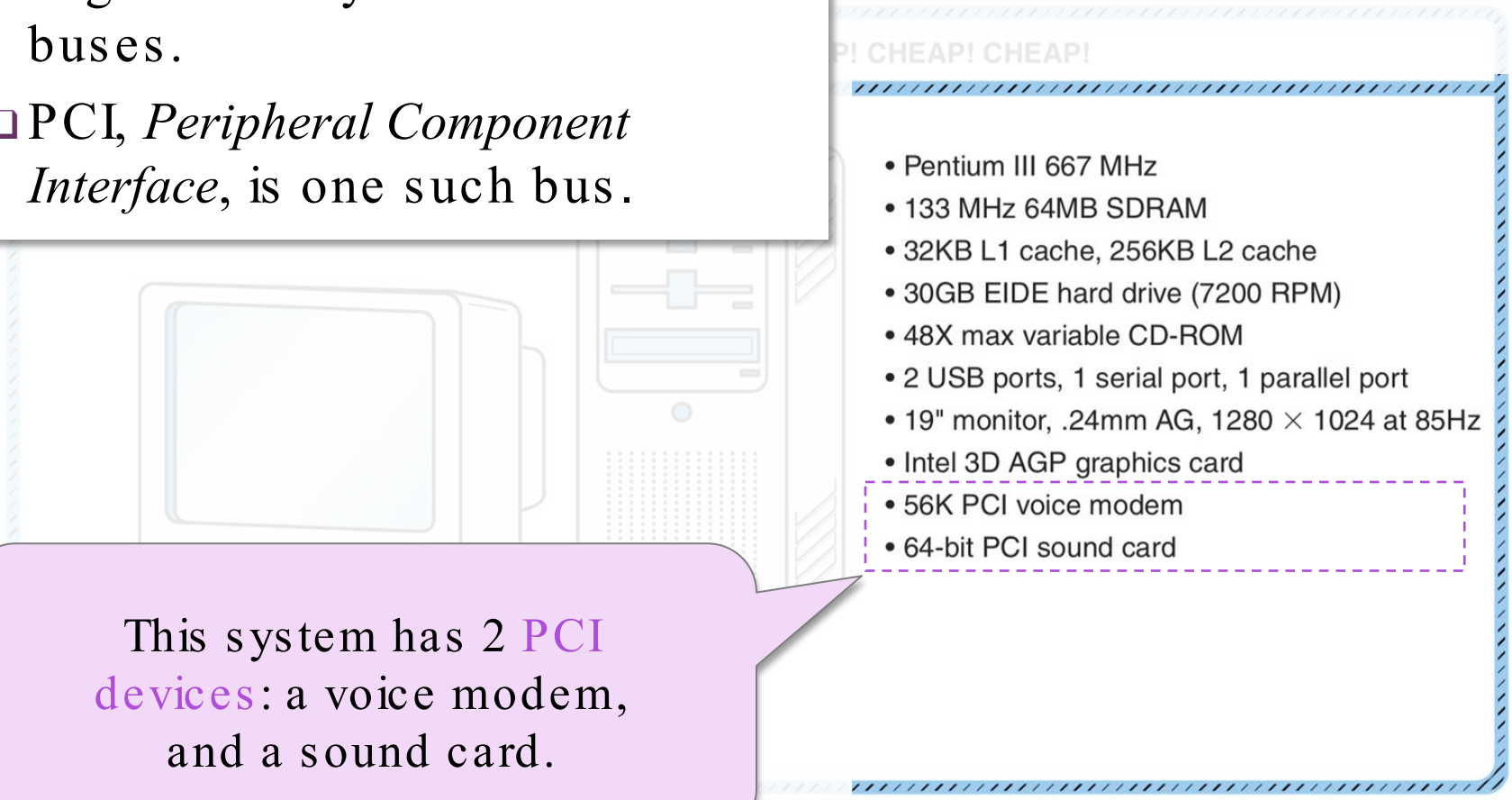
Serial Ports



Parallel Ports

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.9.
https://www.beyondrevelation.co.za/wp-content/uploads/2017/02/Adaptor-15-pin-female-to-male-VGA-INSTAL3.gif
https://cdn.techterms.com/img/lg/parallel_port_105-2.jpg

- System buses can be augmented by dedicated I/O buses.
- PCI, *Peripheral Component Interface*, is one such bus.

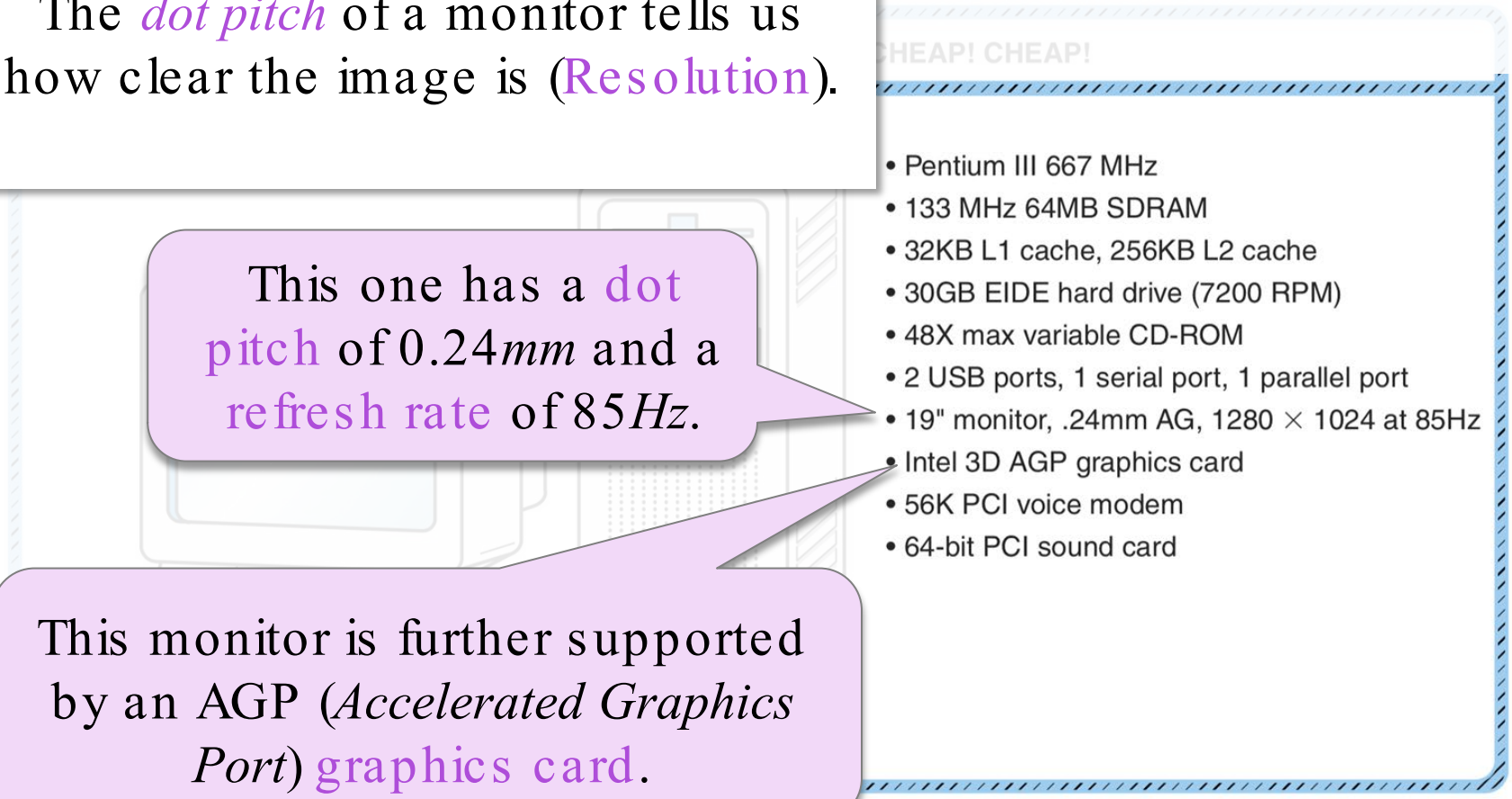


This system has 2 PCI devices: a voice modem, and a sound card.

Figure: A typical computer advertisement.

The number of times per second that the image on a monitor is repainted is its *refresh rate*.

The *dot pitch* of a monitor tells us how clear the image is (Resolution).



This one has a dot pitch of 0.24*mm* and a refresh rate of 85*Hz*.

This monitor is further supported by an AGP (*Accelerated Graphics Port*) graphics card.

Figure : A typical computer advertisement.

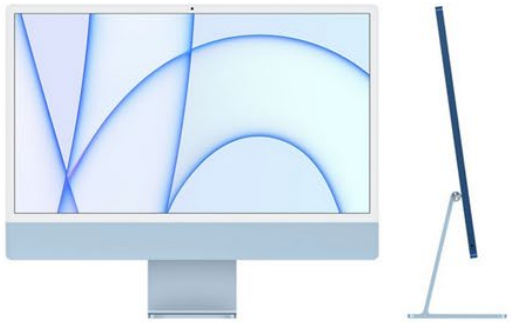
An Example System – Recent ones …



Recent One … mid-range level

Buy the all-new 24″ iMac.

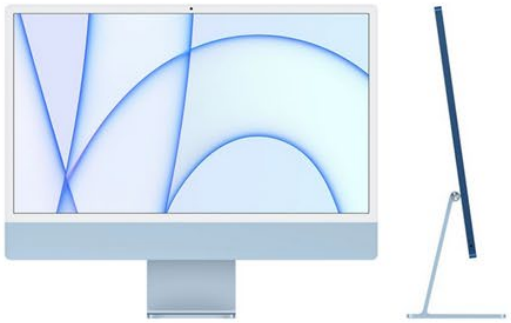
Have questions about buying a Mac?
Chat with a Specialist



Blue

M1 Apple M1 Chip

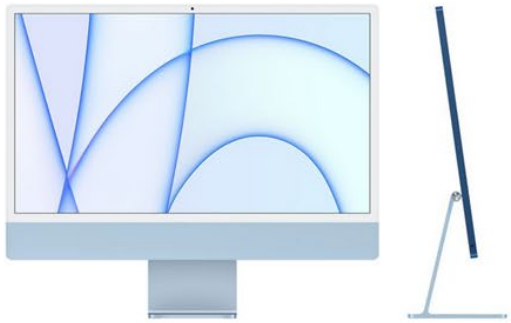
8-Core CPU
7-Core GPU



Blue

M1 Apple M1 Chip

8-Core CPU
8-Core GPU



Blue

M1 Apple M1 Chip

8-Core CPU
8-Core GPU

Recent One … mid-range level

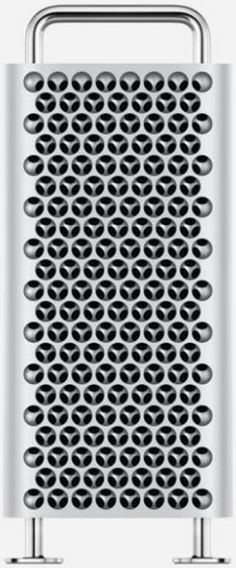
1

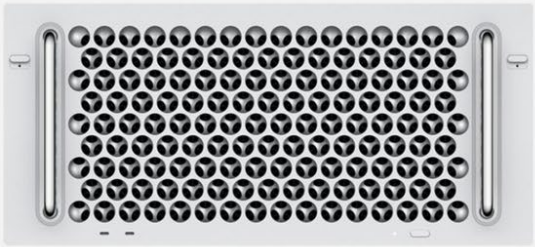
Apple M1 Chip	Apple M1 Chip	Apple M1 Chip
8-Core CPU **7-Core GPU**	**8-Core CPU** **8-Core GPU**	**8-Core CPU** **8-Core GPU**
256GB storage[1]	256GB storage[1]	512GB storage[1]
8GB unified memory	8GB unified memory	8GB unified memory
24-inch 4.5K Retina display[2]	24-inch 4.5K Retina display[2]	24-inch 4.5K Retina display[2]
Two Thunderbolt / USB 4 ports	Two Thunderbolt / USB 4 ports	Two Thunderbolt / USB 4 ports
—	Two USB 3 ports	Two USB 3 ports
—	Gigabit Ethernet	Gigabit Ethernet
Magic Keyboard	Magic Keyboard with Touch ID	Magic Keyboard with Touch ID
RM 5,599.00	**RM 6,449.00**	**RM 7,249.00**
Select	Select	Select

Recent One … high-end level

1

Choose your Mac Pro.





Mac Pro

Tower from RM 25,999.00

Configure

Mac Pro

Rack from RM 27,999.00

Configure

Recent One … high-end level



Mac Pro

Processor Configure from an 8-core to 28-core Intel Xeon W processor

8-Core

3.5GHz Intel Xeon W

8 cores, 16 threads

Turbo Boost up to 4.0GHz

24.5MB cache

Supports 2666MHz memory

Recent One … high-end level

1

Mac Pro

12-Core

3.3GHz Intel Xeon W

12 cores, 24 threads

Turbo Boost up to 4.4GHz

31.25MB cache

Supports 2933MHz memory

16-Core

3.2GHz Intel Xeon W

16 cores, 32 threads

Turbo Boost up to 4.4GHz

38MB cache

Supports 2933MHz memory

Mac Pro

24-Core

2.7GHz Intel Xeon W

24 cores, 48 threads

Turbo Boost up to 4.4GHz

57MB cache

Supports 2933MHz memory

28-Core

2.5GHz Intel Xeon W

28 cores, 56 threads

Turbo Boost up to 4.4GHz

66.5MB cache

Supports 2933MHz memory

1.7 Summary

1

- The distinction between computer organization and computer architecture is not clear-cut.
- In fact, neither computer organization nor computer architecture can stand alone. They are interrelated and interdependent.
- Students can truly understand each of these fields only after they comprehend both throughout the remainder of this course.
- This leads to a deeper understanding of computers and computation — the heart and soul of computer science.

Review Questions

1

1.1 What, in general terms, is the distinction between computer organization and computer architecture?

1.2 What, in general terms, is the distinction between computer structure and computer function?

1.3 What are the four main functions of a computer?

1.4 List and briefly define the main structural components of a computer.

1.5 List and briefly define the main structural components of a processor.

Review Questions

1

1.6 What is a stored program computer?

1.7 What are the four main components of any general-purpose computer?

1.8 List and explain the key characteristics of a computer family.

1.9 What is the key distinguishing feature of a microprocessor?